Faculty Development Program

3rd day (2024/06/26)

Seung Hyeon Yang
Research Professor, College of Nursing Science,
Kyung Hee University, Republic of Korea

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1. Introduction

5 Main Goals of the BK-LUPIC Project

1) Reform of educational programs in SD;
2) Improvement of the educational environment;
3) Strengthening the capacity of teaching staff;
4) Strengthening the capacity of medical professionals in the field of medical information processing;
5) Improvement of the e-health system

These programs

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1. Introduction

학습목표

연구능력 향상을 위하여 통계 프로그램(JASP) 사용방법을 이해하고, 이를 사용하여 데이터를 분석한다.

JASP

https://jasp-stats.org/

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1. Introduction

오늘 강의

≠ 통계학 강의

= 통계프로그램 강의

JASP

https://jasp-stats.org/

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2. JASP

특징

1. R 기반 통계 소프트웨어
   1) 무료
   2) 확장성
2. 사용자 친화적 인터페이스
3. 러시아어 지원

JASP

https://jasp-stats.org/

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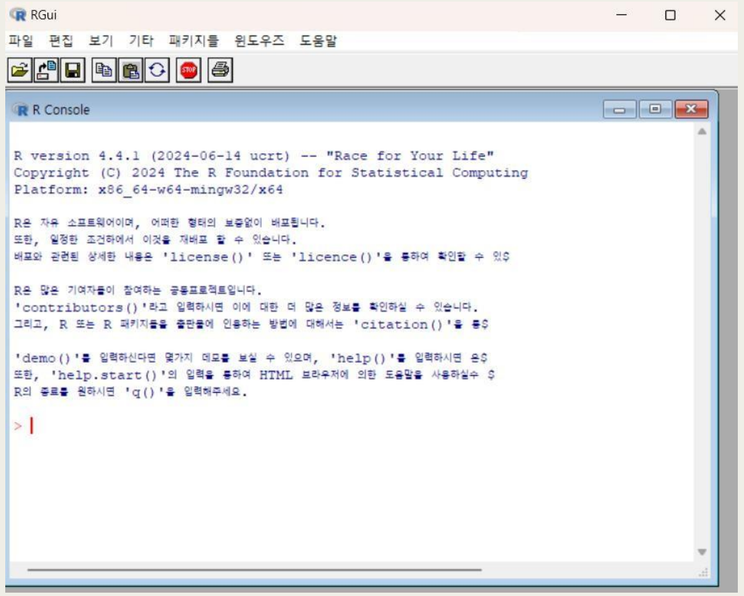
2. JASP



장점	단점
무료	
	명령문 기반
다양한 통계 패키지 (Powerful)	패키지 신뢰성
	패키지 검색 및 설치

Subscription

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Save 10% on annual subscription

Annual or monthly subscriptions. Extend the Base subscription with three optional add-ons.

Base	Standard	Professional	Premium
USD 3,830 per user**	**USD 8,440 per user****	**USD 16,900 per user****	**USD 25,200 per user****
A perpetual or term license with SPSS Statistics Base features	A perpetual or term license with SPSS Statistics Base and Custom Tables and Advanced Statistics modules	A perpetual or term license of SPSS Statistics including a large selection of additional capabilities	A perpetual or term license including the full catalog of SPSS Statistics modules and capabilities

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2. JASP

장점	단점
무료	
	명령문 기반
다양한 통계 패키지 (Powerful)	패키지 신뢰성
	패키지 검색 및 설치

장점	단점
무료	
마우스 클릭 기반	~~명령문 기반~~
다양한 통계 패키지 (Powerful)	~~패키지 신뢰성~~
	~~패키지 검색 및 설치~~

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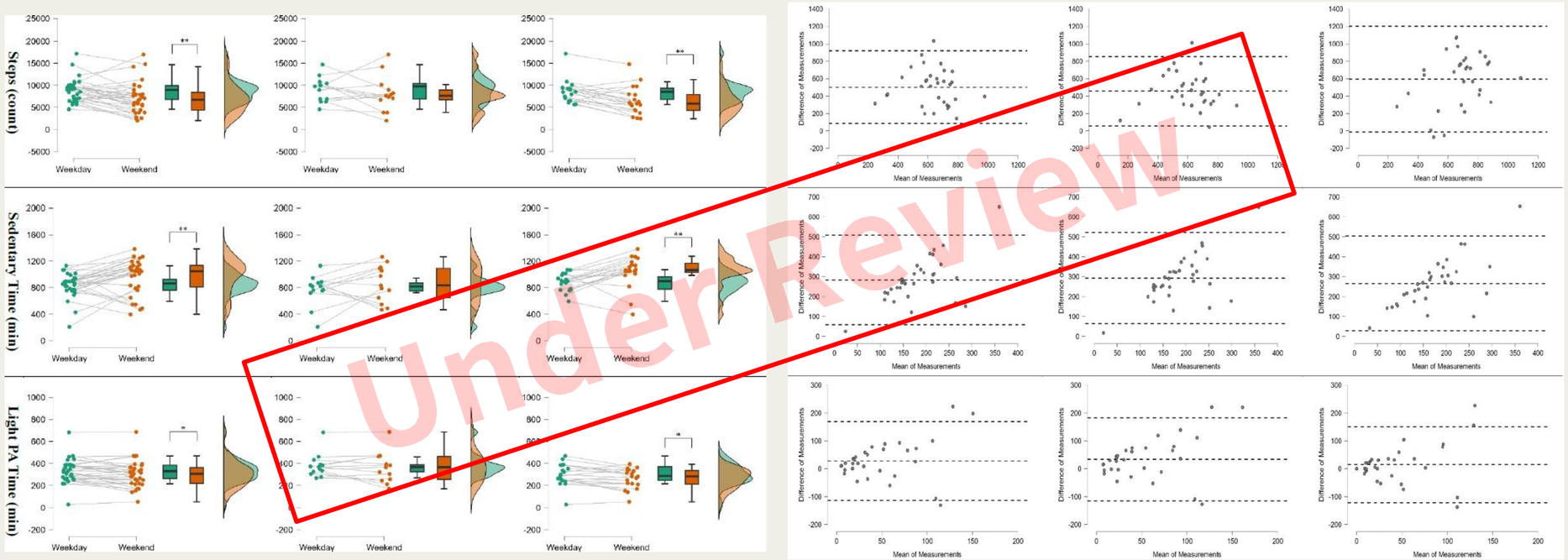
2. JASP

JASP



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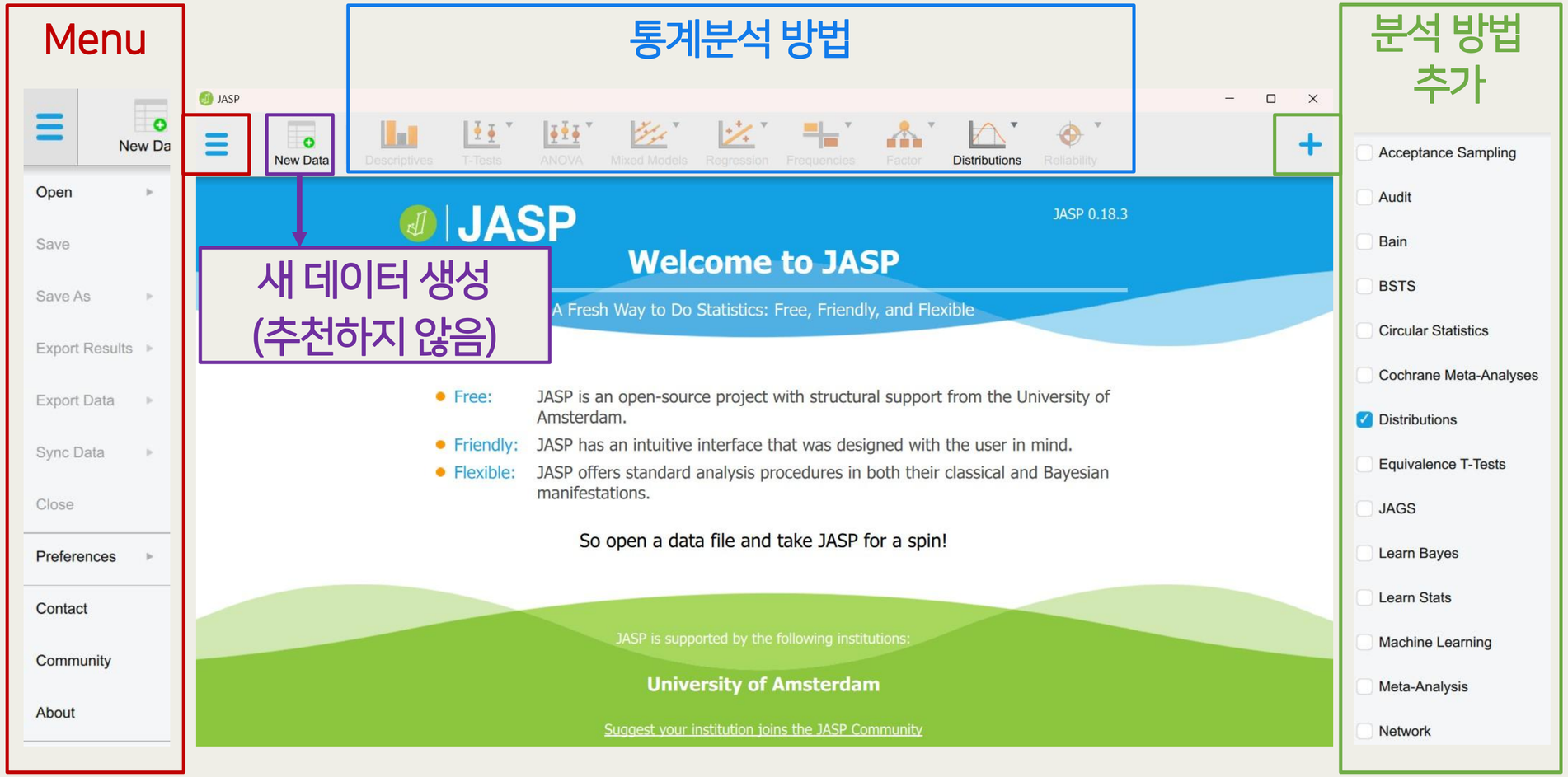
3. Interface



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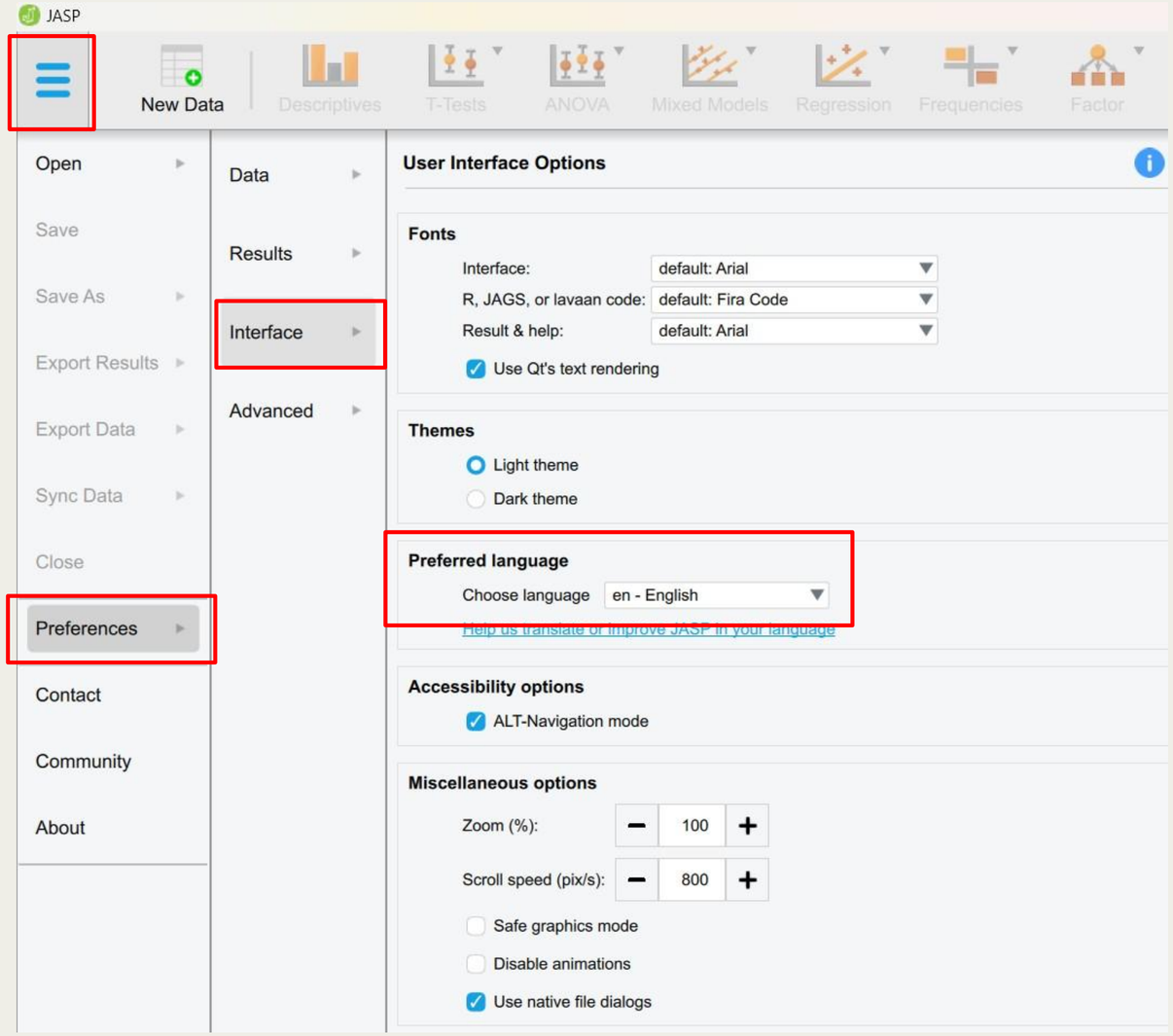
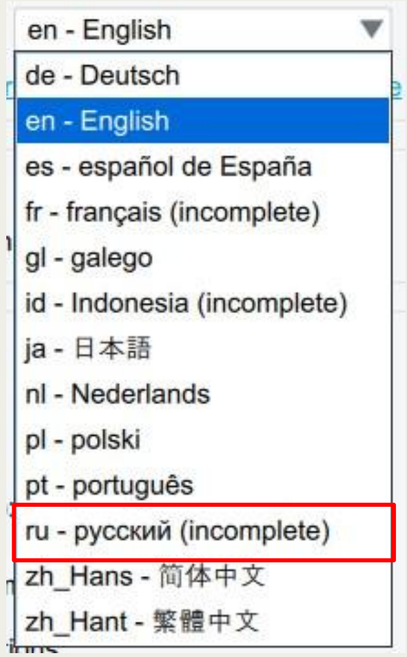
3. Interface





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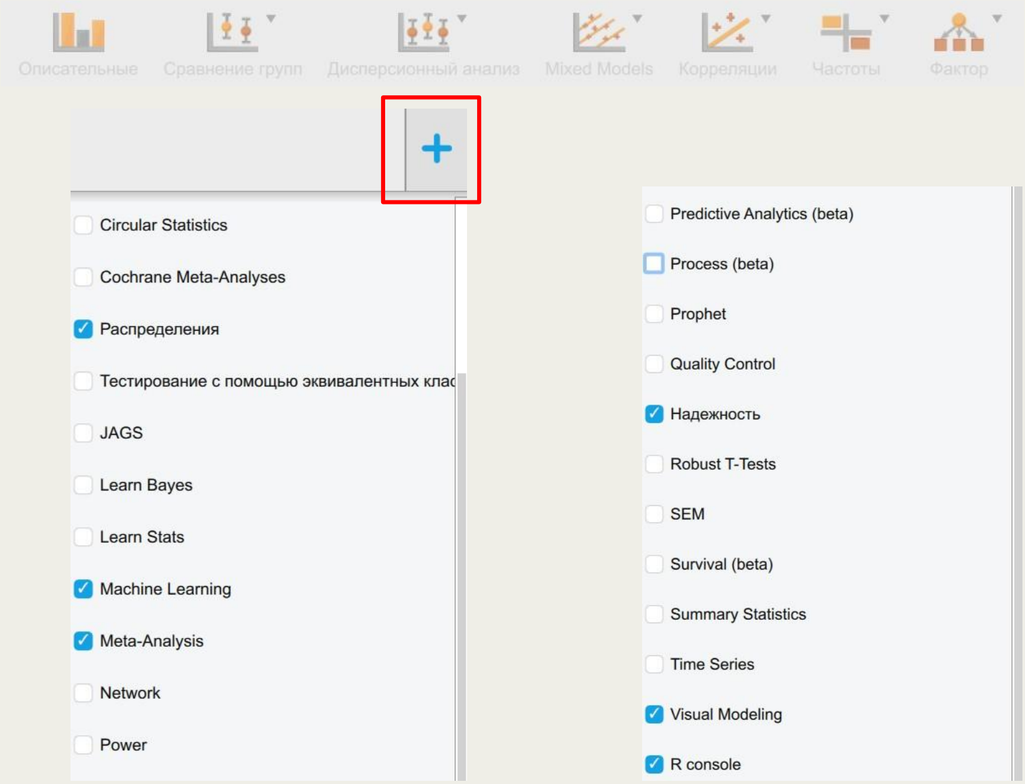
3. Interface



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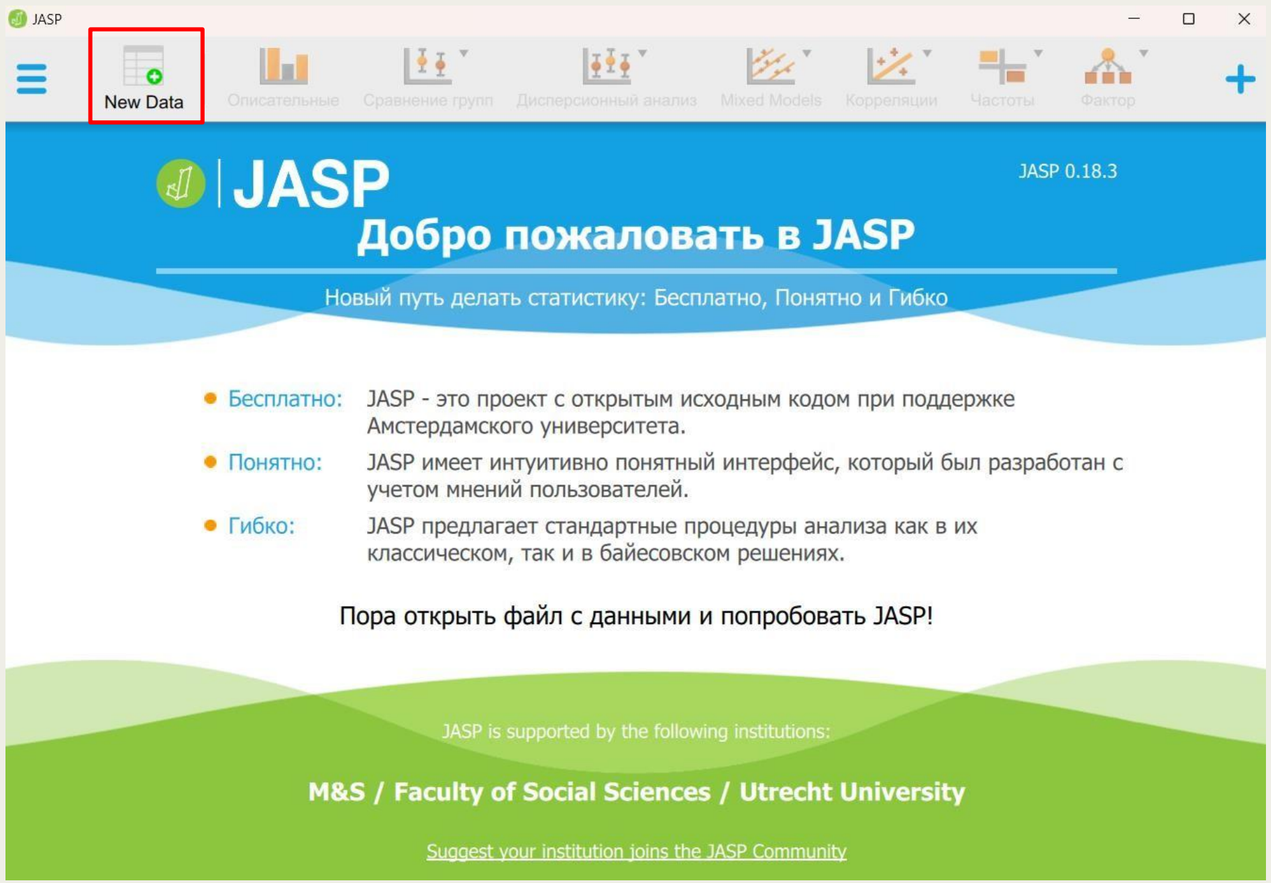

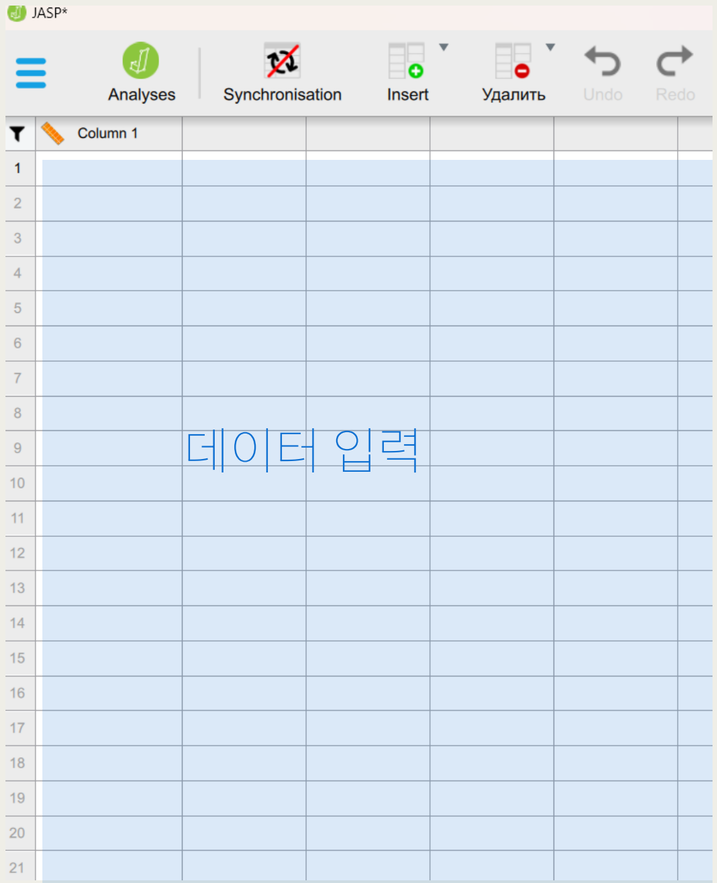

추천하지 않음

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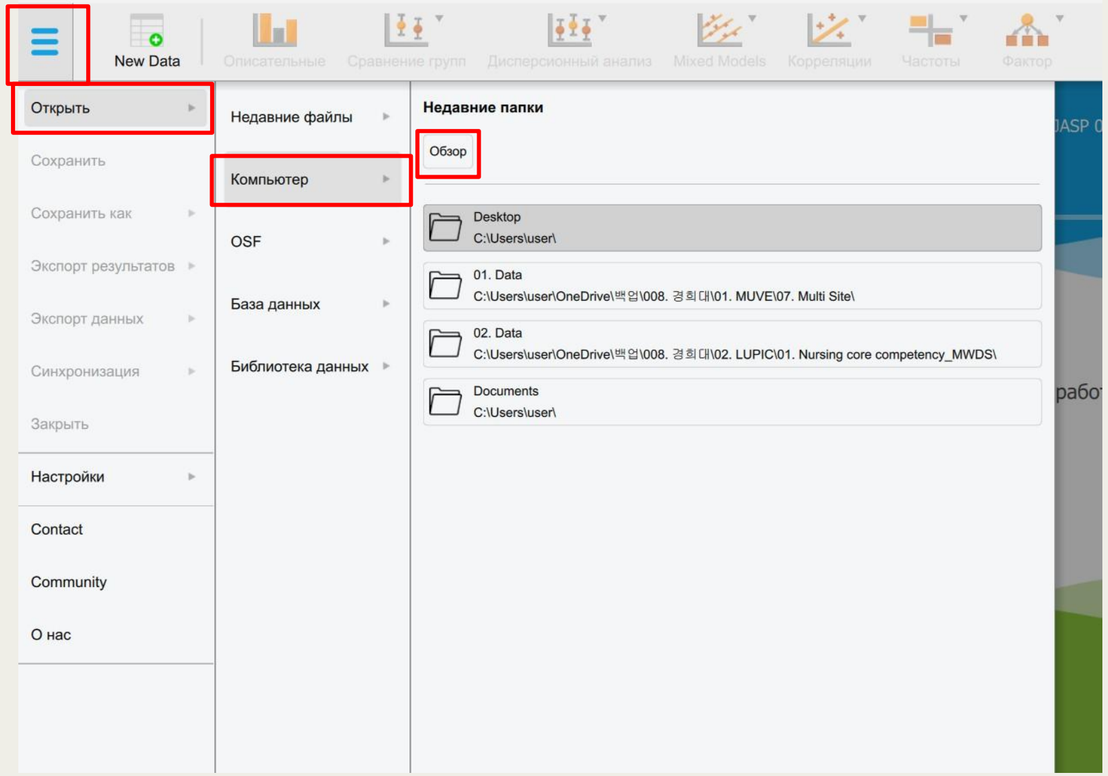
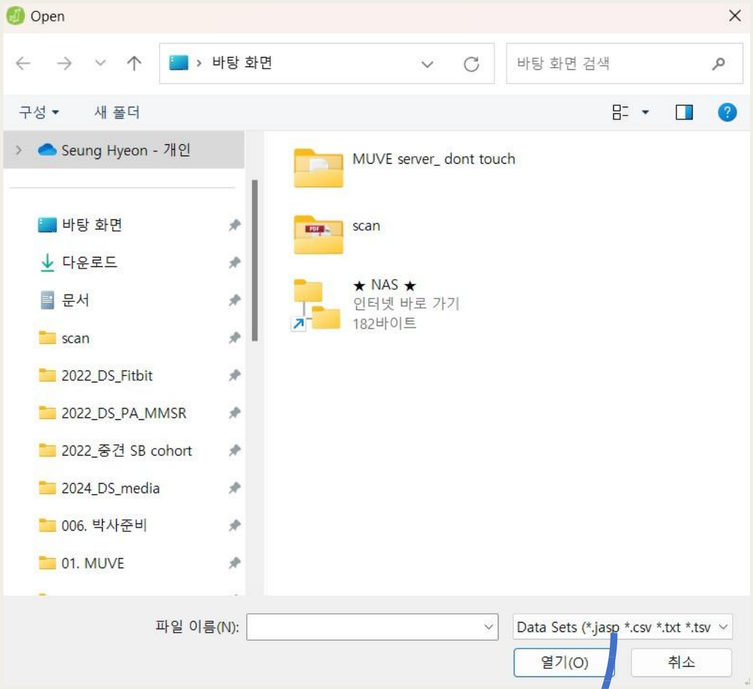
4. Data - Open





Data Sets (*.jasp *.csv *.txt *.tsv *.sav *.zsav *.ods *.dta *.por *.sas7bdat *.sas7bcat *.xpt)

!! No !! *.xlsx / *.xls

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4. Data - Open

.xlsx / .csv

Особенности	xlsx	csv (Значения, разделенные запятыми)
Лист	Несколько листов	Один лист
Совместимость	MS excel и другие совместимые программы	Текстовый редактор и программа для работы с электронными таблицами
Поддержка форматирования	Поддерживает различные элементы, такие как форматирование, диаграммы, изображения и т.д.	Не поддерживает форматирование, диаграммы, изображения и т.д.
Функция	O	X
Макрос	O	X
Редактирование	Легко	Сложно
Цель использования	Анализирование сложных данных, создание отчетов и т.д.	Обмен и хранение данных

SPSS

SPSS & JASP

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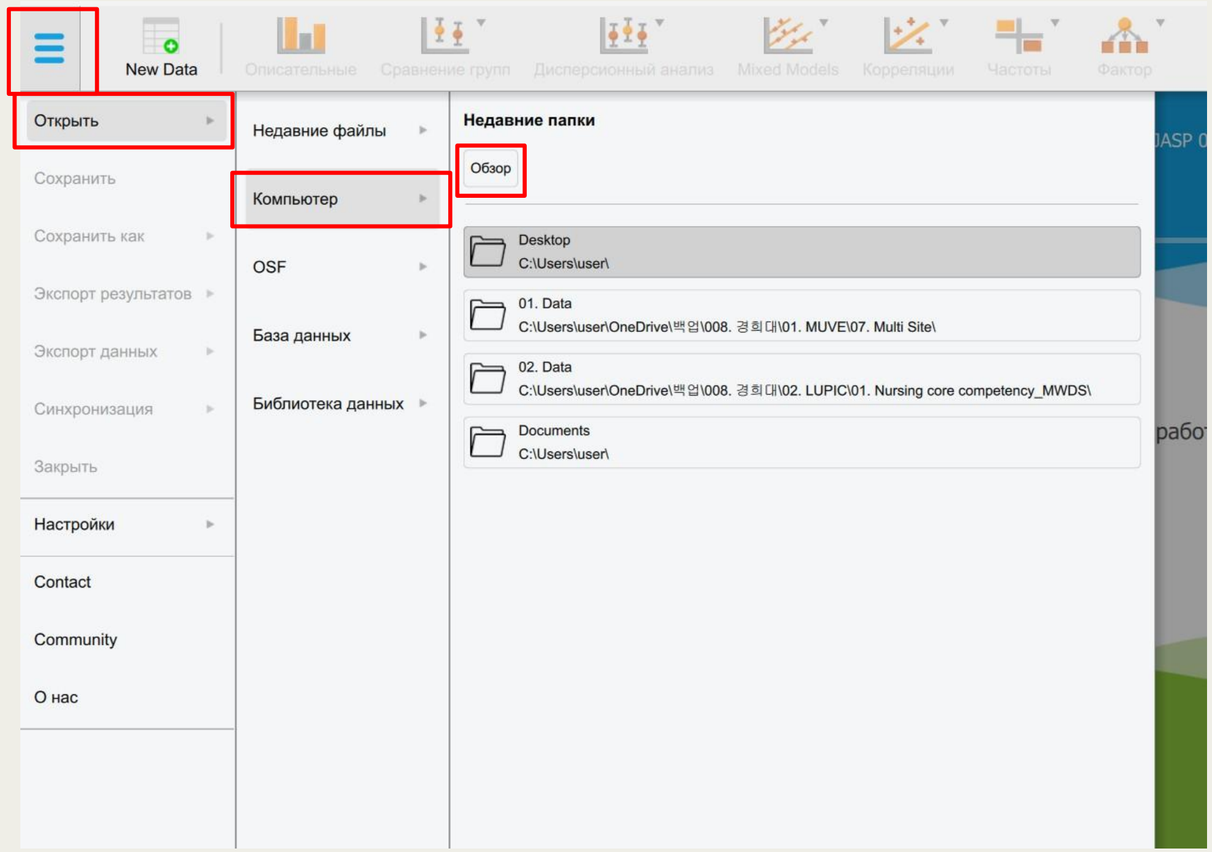
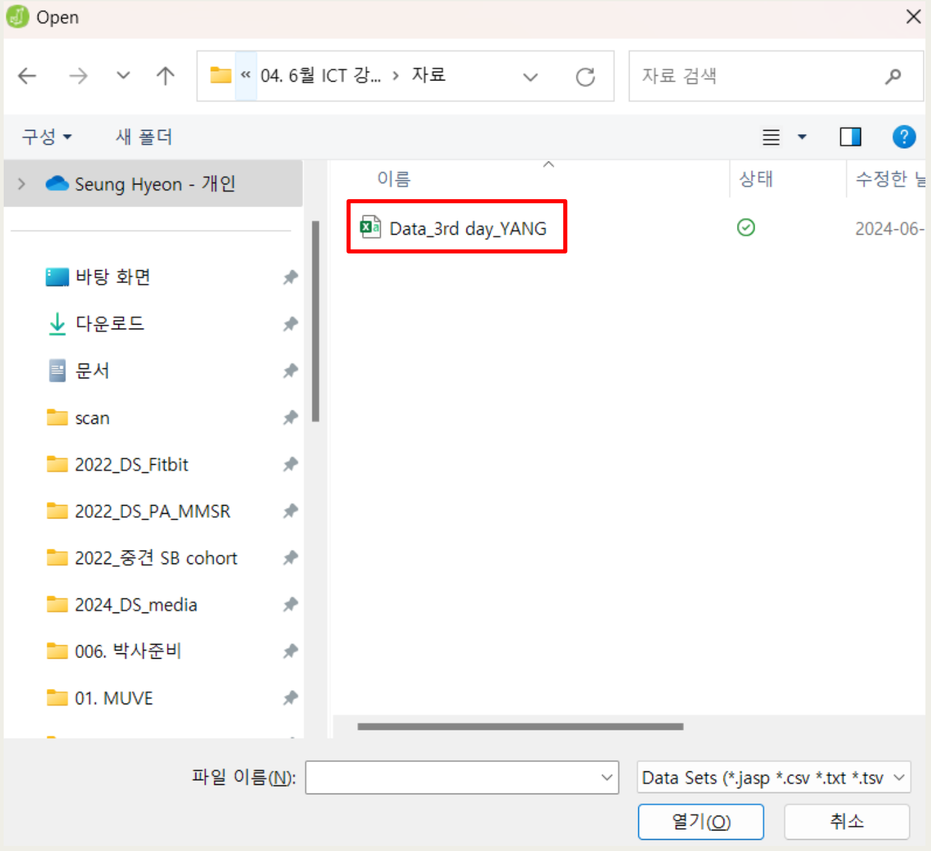
4. Data - Open





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4. Data - Open

Data_3rd day_YANG (C:\Users\user\OneDrive\백업\008. 경희대\02. LUPIC\04. 6월 ICT 강의\자료)

Edit Data | Описательные | Сравнение групп | Дисперсионный анализ | Mixed Models | Корреляции | Частоты | Фактор

	No	Age	Sex	School	Score_pre	Score_post
1	1	22	2	1	60	60
2	2	23	2	1	48	57
3	3	22	2	1	66	80
4	4	22	2	1	58	73
5	5	24	1	1	61	63
6	6	24	1	1	60	60
7	7	23	2	1	79	80
8	8	24	1	1	69	75
9	9	24	2	1	51	60
10	10	22	1	1	61	67
11	11	22	2	1	80	80
12	12	24	1	1	64	80
13	13	22	2	1	54	73
14	14	22	2	1	70	78
15	15	22	2	1	80	80
16	16	22	2	1	60	66
17	17	22	2	1	68	80
18	18	22	2	1	80	70
19	19	23	2	1	77	80
20	20	24	1	1	80	80
21	21	21	2	1	61	80

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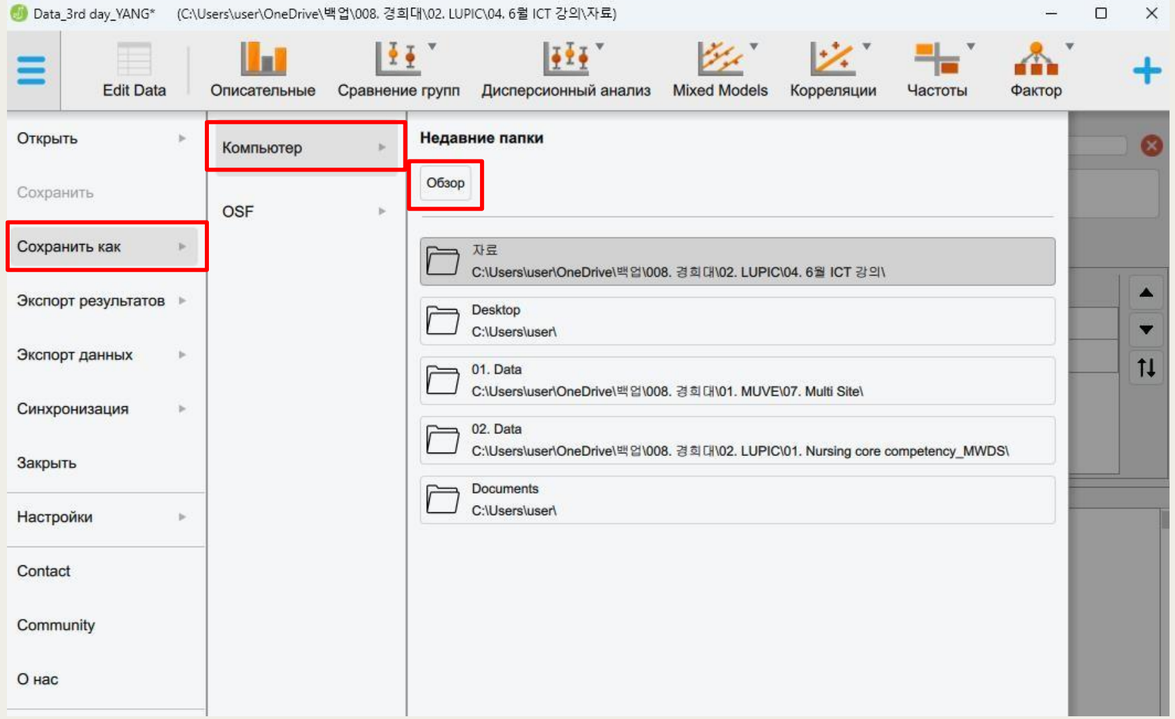
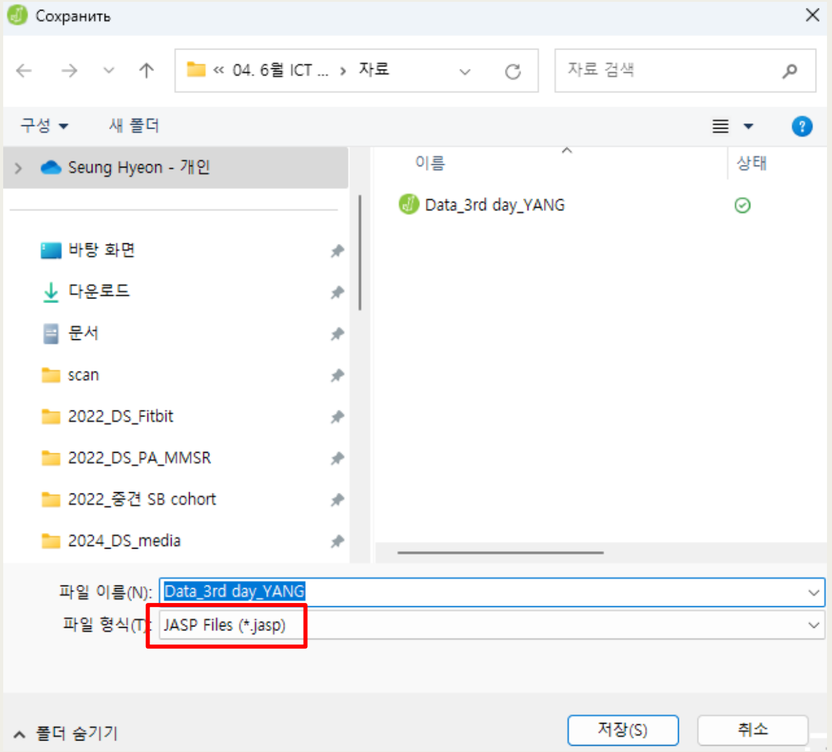
4. Data - Save



or

Ctrl + S



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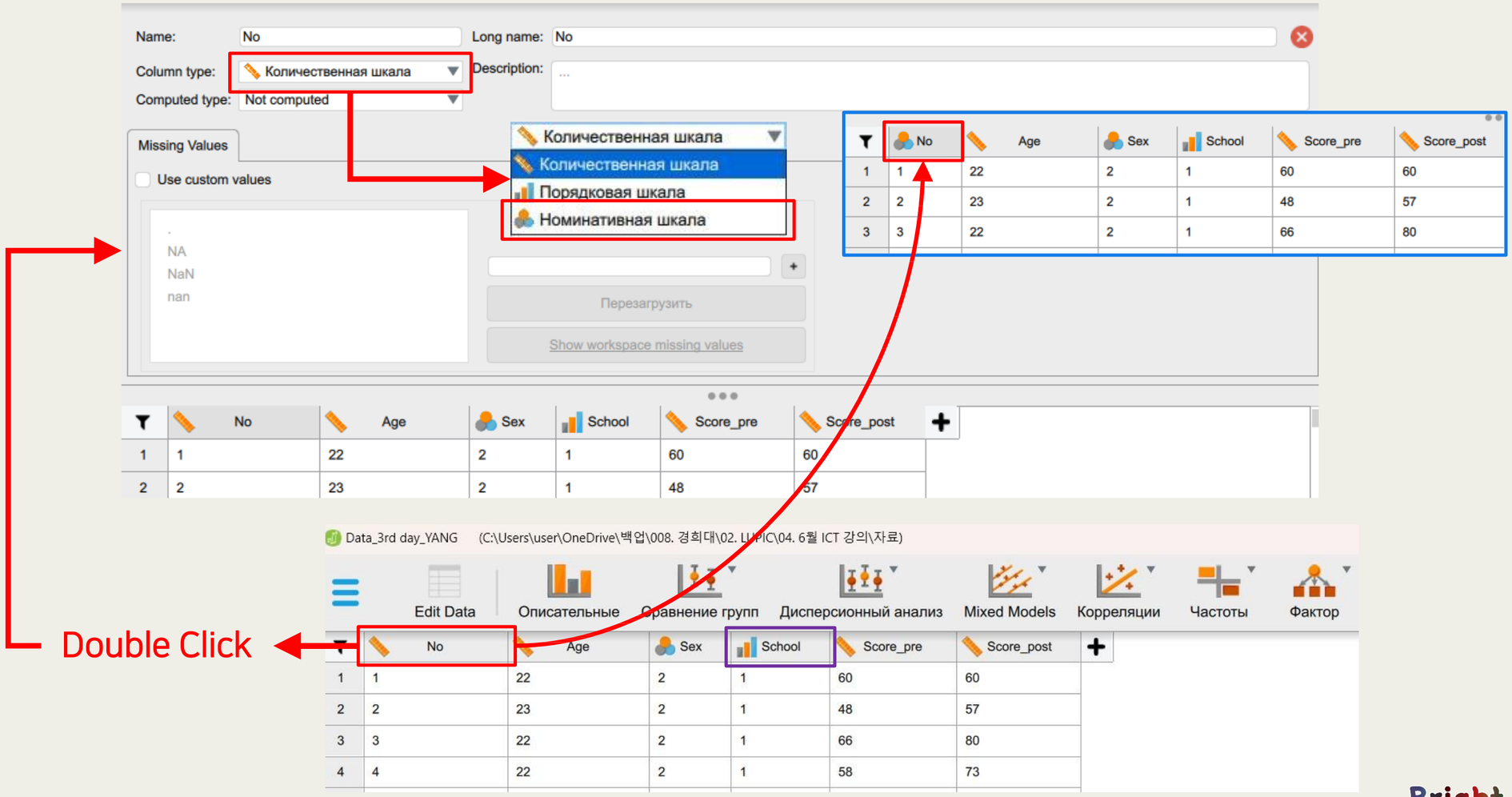
4. Data - Type



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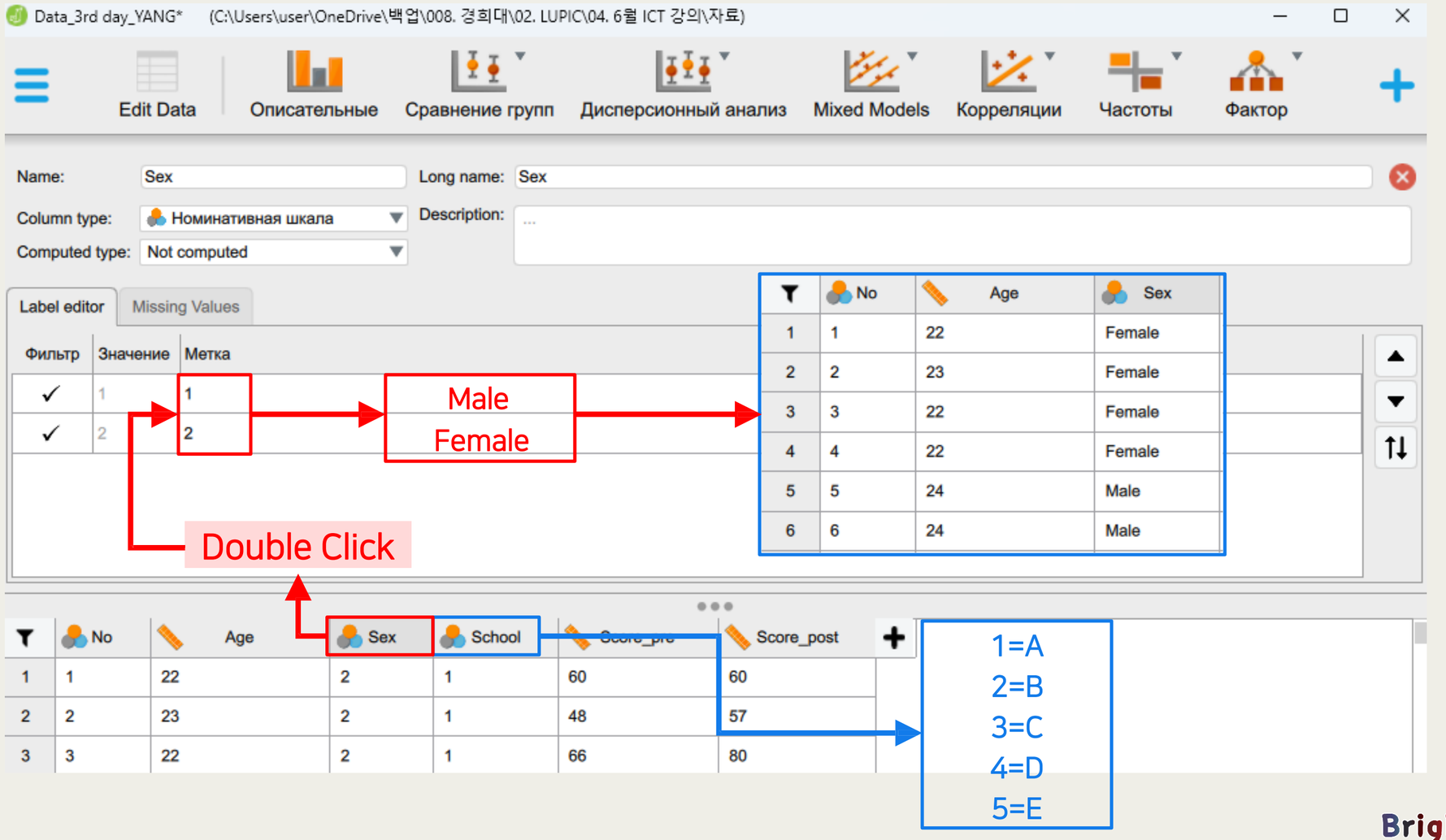
4. Data - Label



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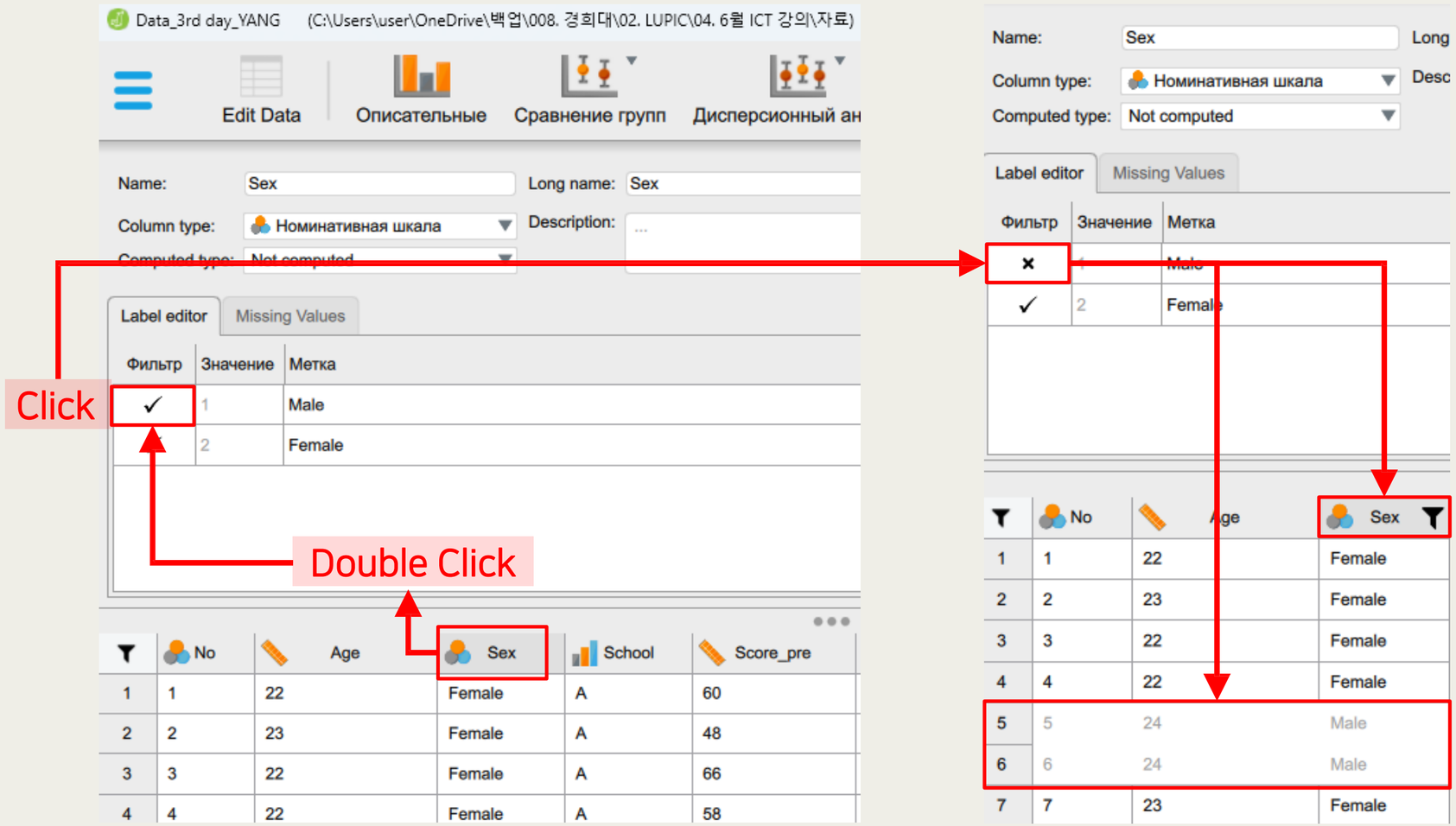
4. Data - Filter



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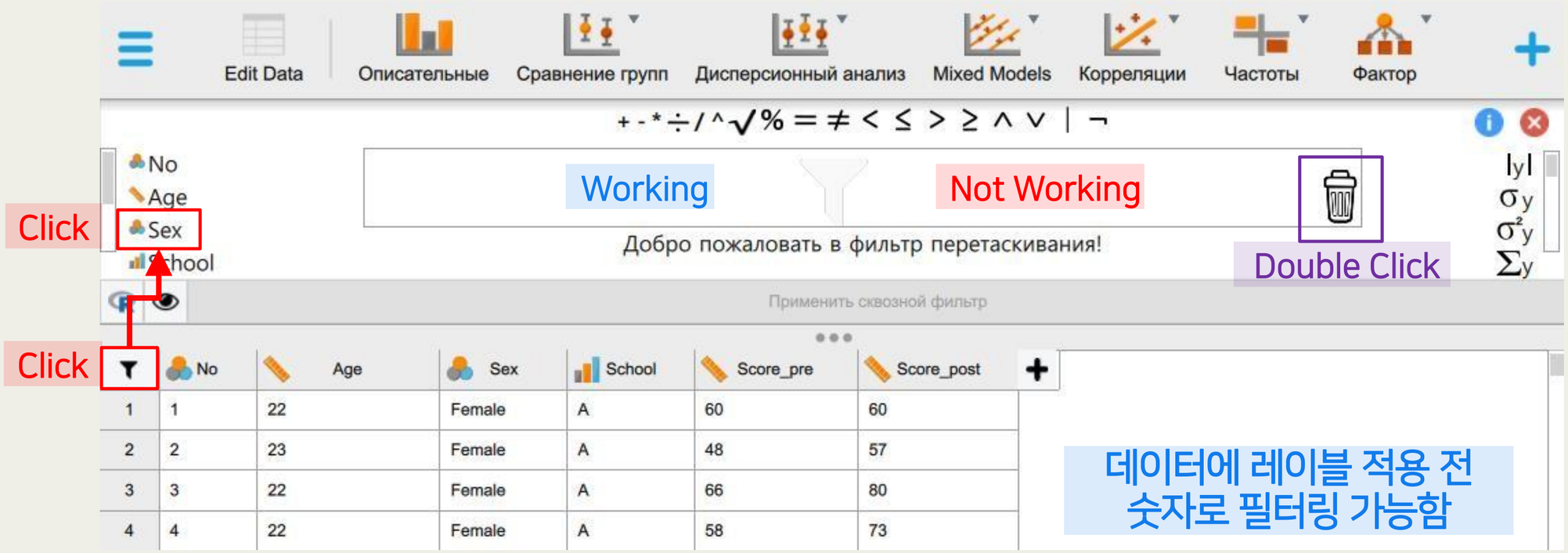

데이터에 레이블 적용 전
숫자로 필터링 가능함

데이터에 레이블 적용 후
숫자로 필터링 불가능함

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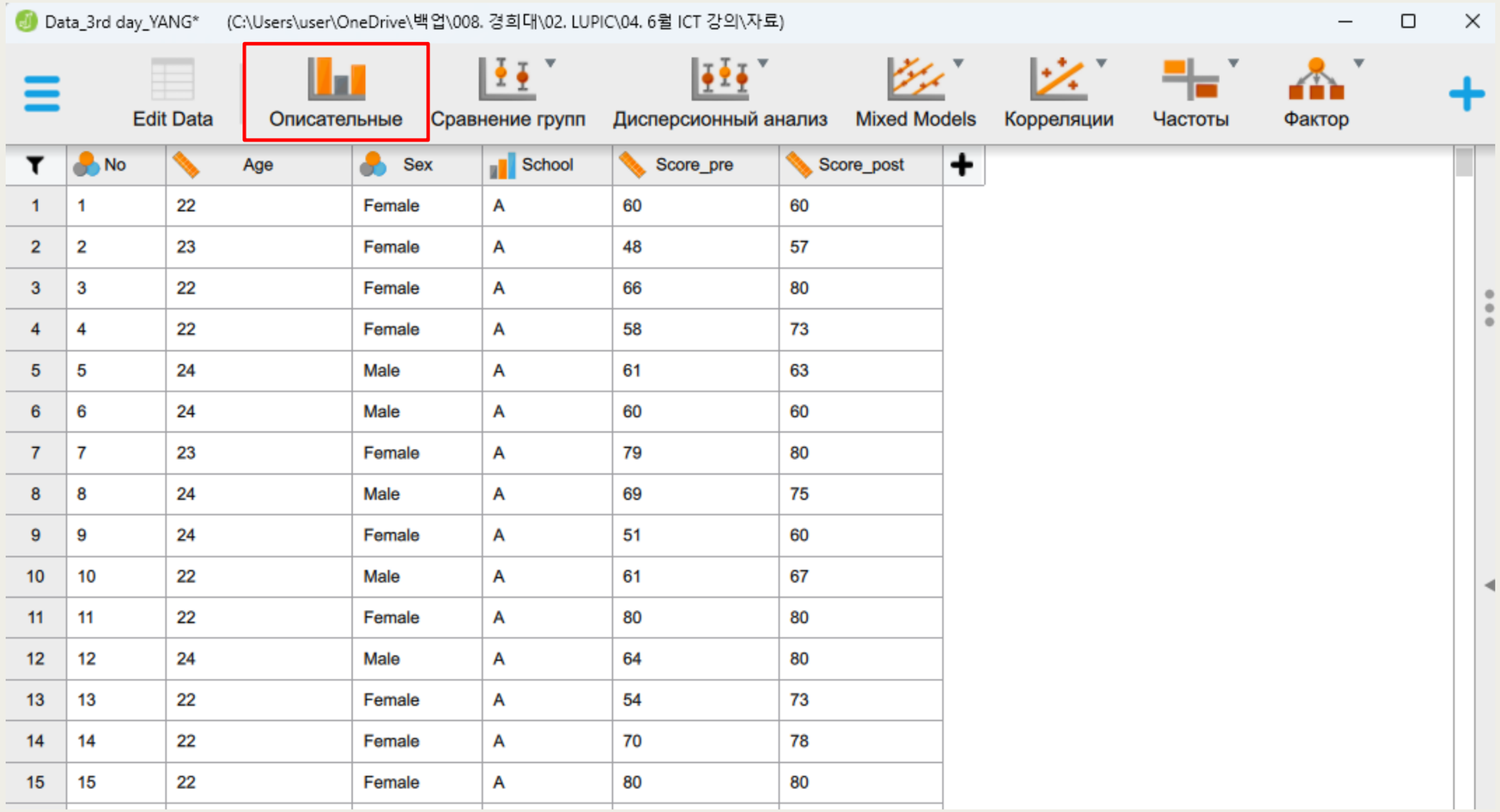
5. Analysis - Layout



	No	Age	Sex	School	Score_pre	Score_post
1	1	22	Female	A	60	60
2	2	23	Female	A	48	57
3	3	22	Female	A	66	80
4	4	22	Female	A	58	73
5	5	24	Male	A	61	63
6	6	24	Male	A	60	60
7	7	23	Female	A	79	80
8	8	24	Male	A	69	75
9	9	24	Female	A	51	60
10	10	22	Male	A	61	67
11	11	22	Female	A	80	80
12	12	24	Male	A	64	80
13	13	22	Female	A	54	73
14	14	22	Female	A	70	78
15	15	22	Female	A	80	80

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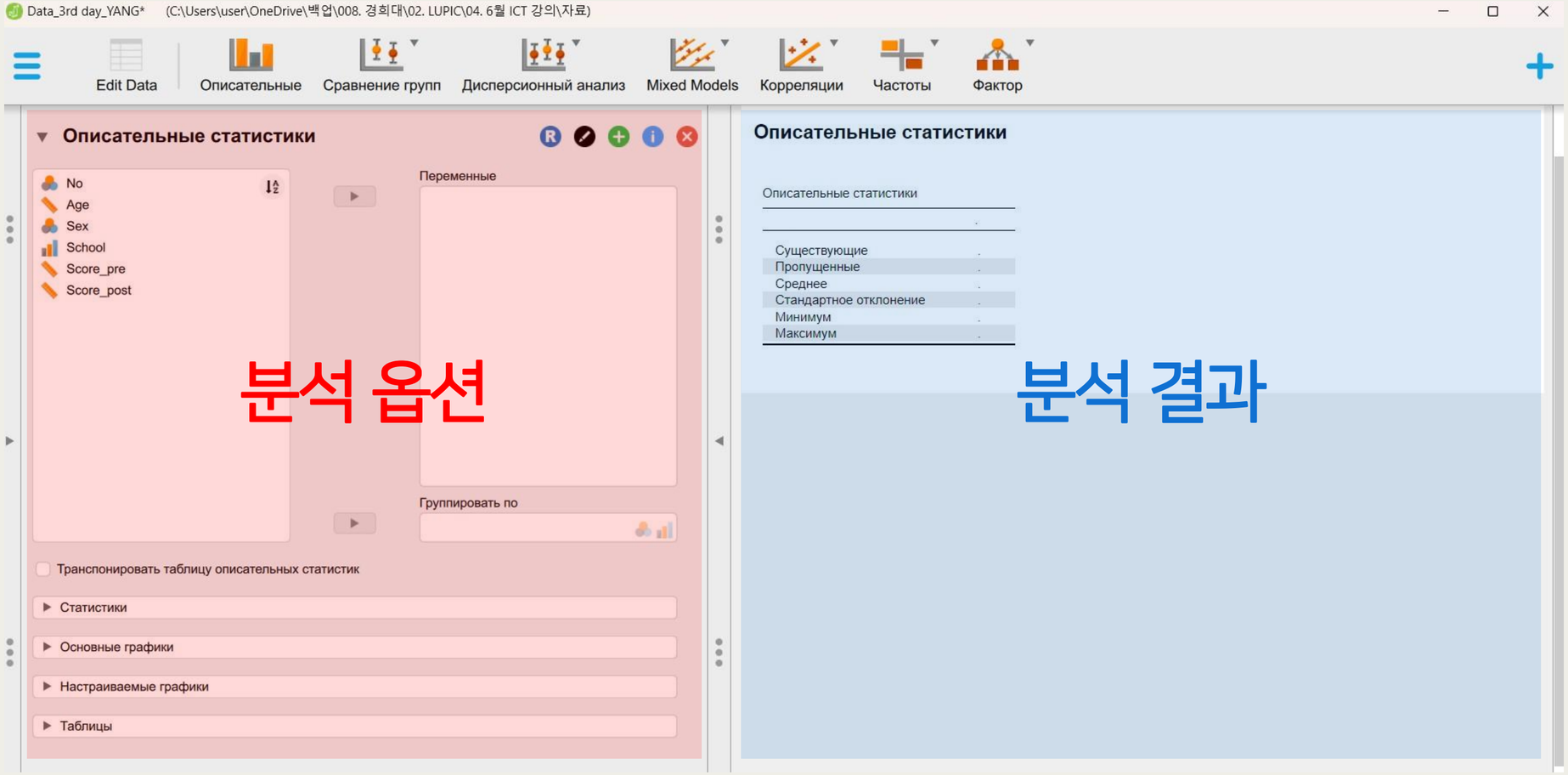
5. Analysis - Layout



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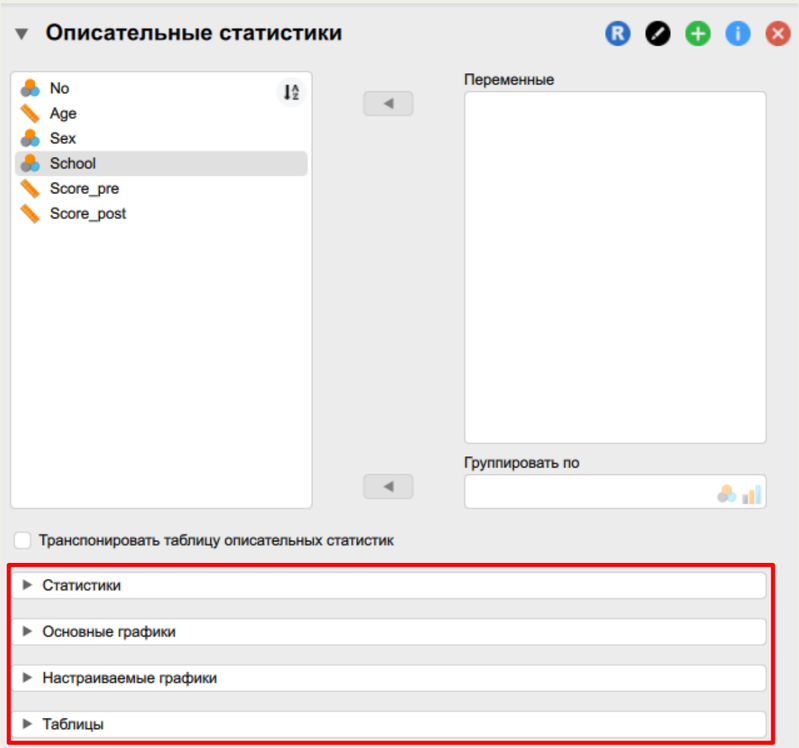

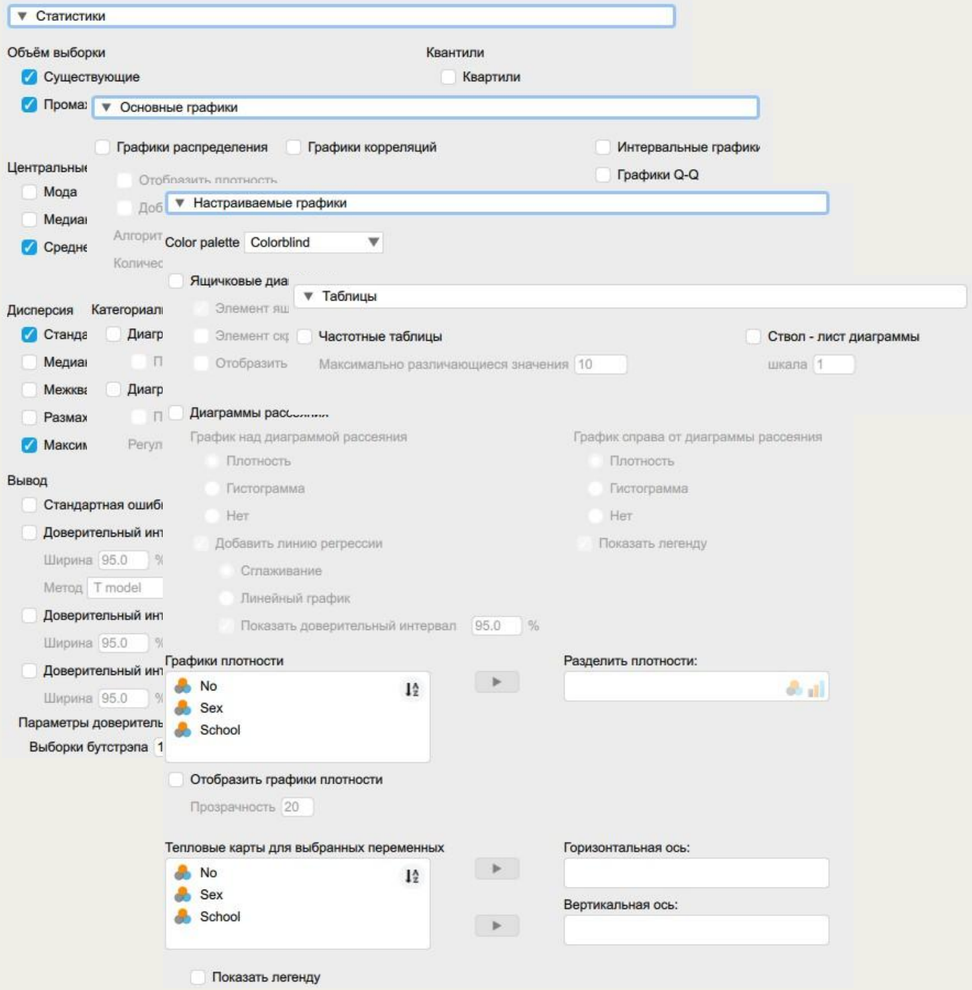

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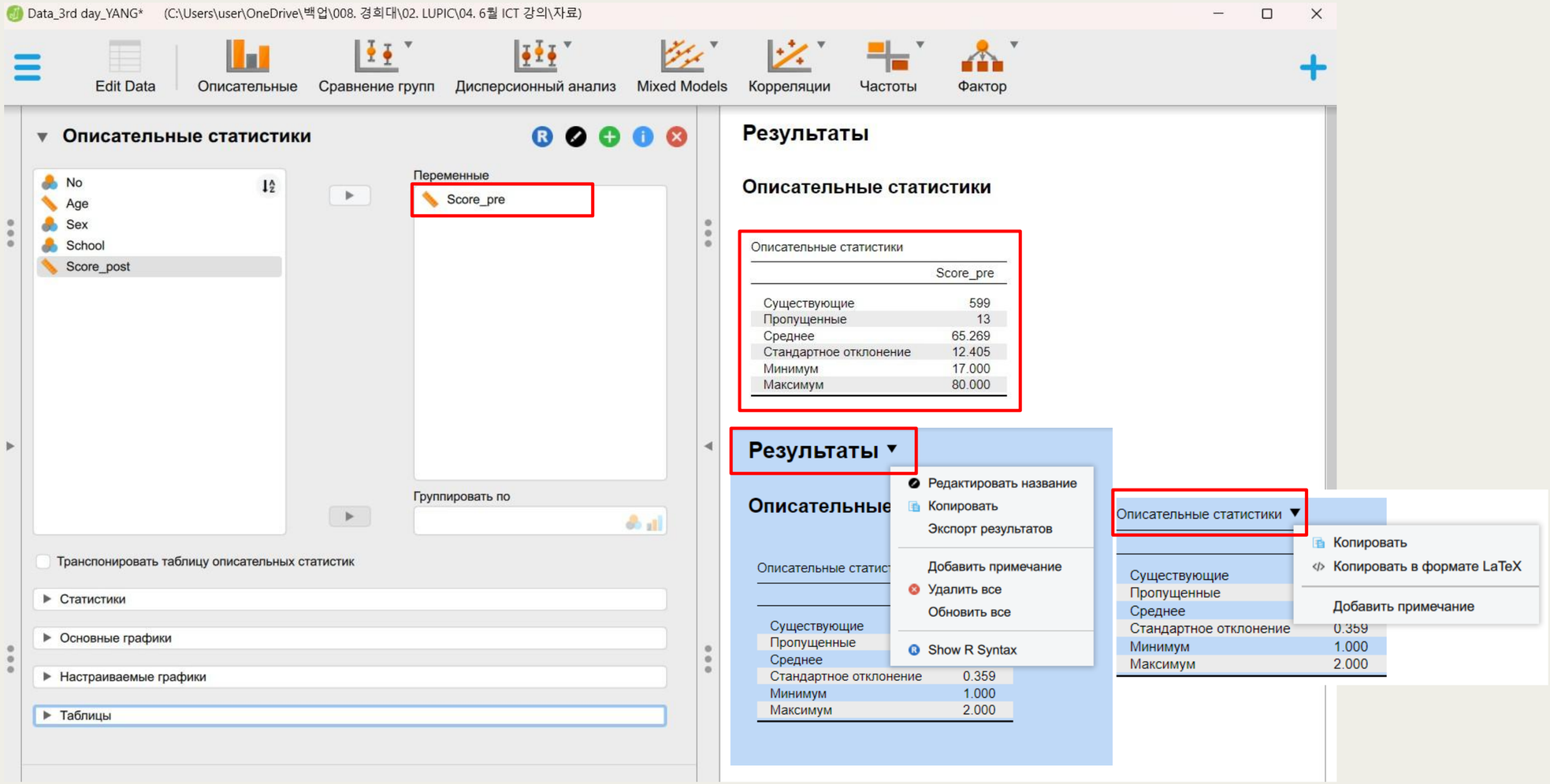
6. Descriptive



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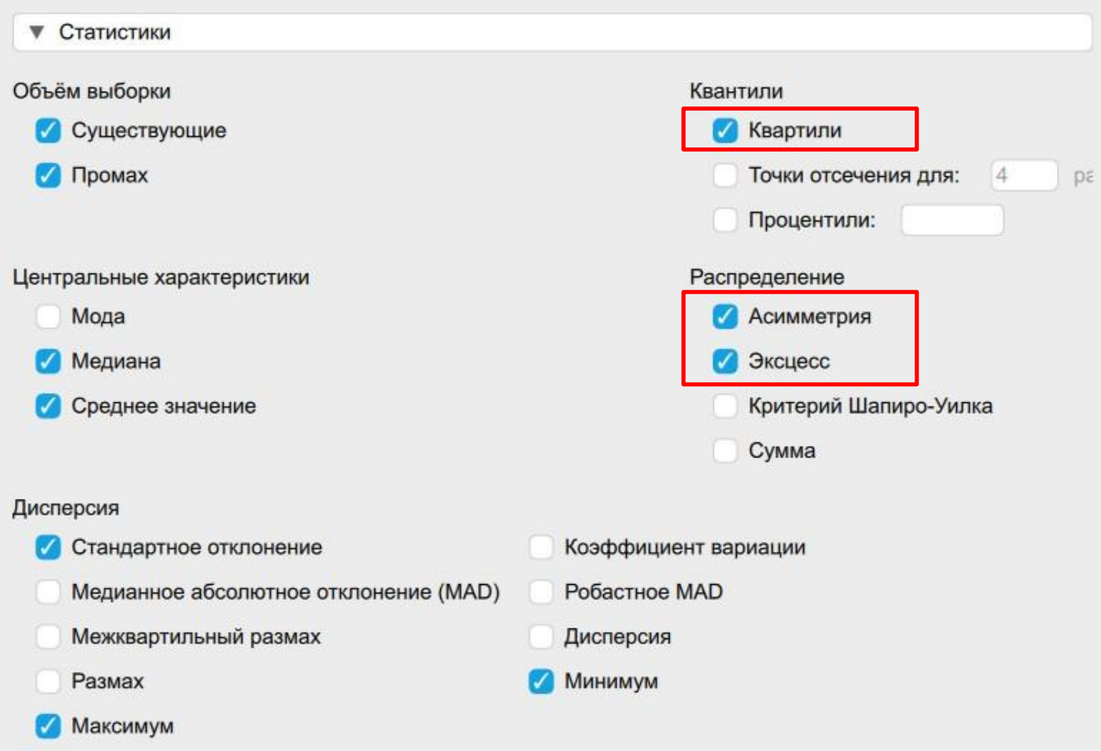
6. Descriptive



Описательные статистики

	Score_pre
Существующие	599
Пропущенные	13
Медиана	63.000
Среднее	65.269
Стандартное отклонение	12.405
Асимметрия	−0.506
Стандартная ошибка асимметрии	0.100
Эксцесс	−0.184
Стандартная ошибка эксцесса	0.199
Минимум	17.000
Максимум	80.000
25-й процентиль	59.000
50-й процентиль	63.000
75-й процентиль	79.000

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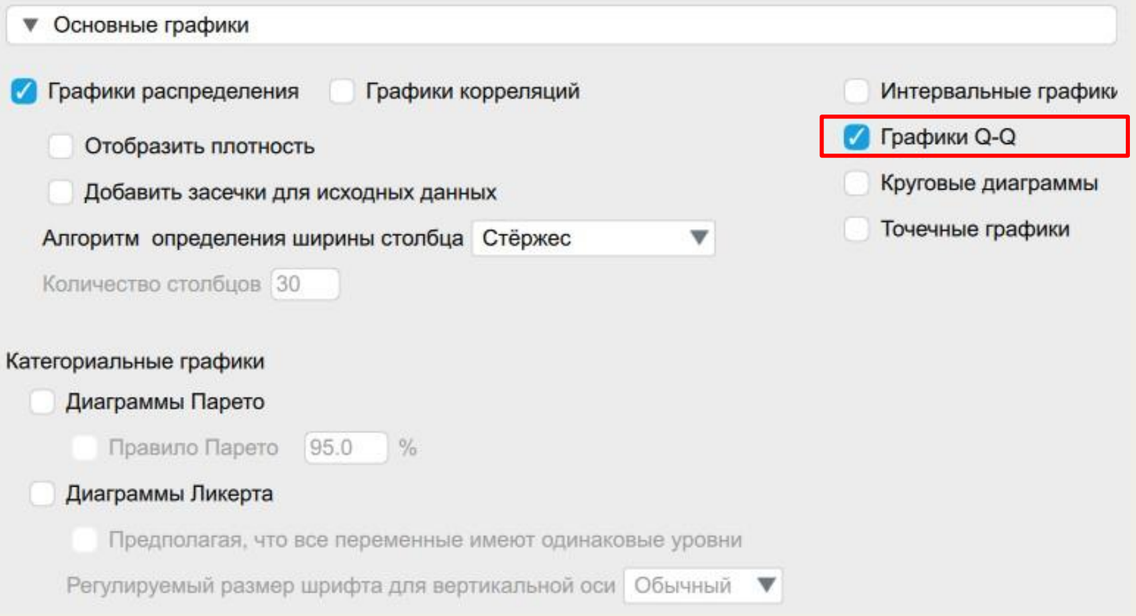
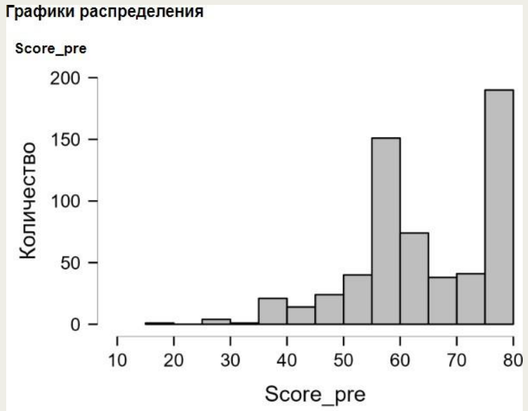
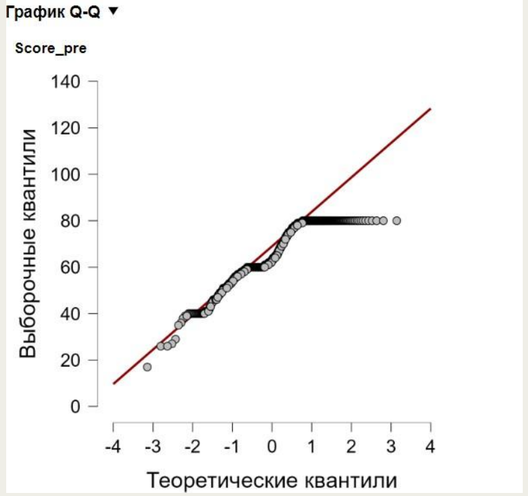
6. Descriptive







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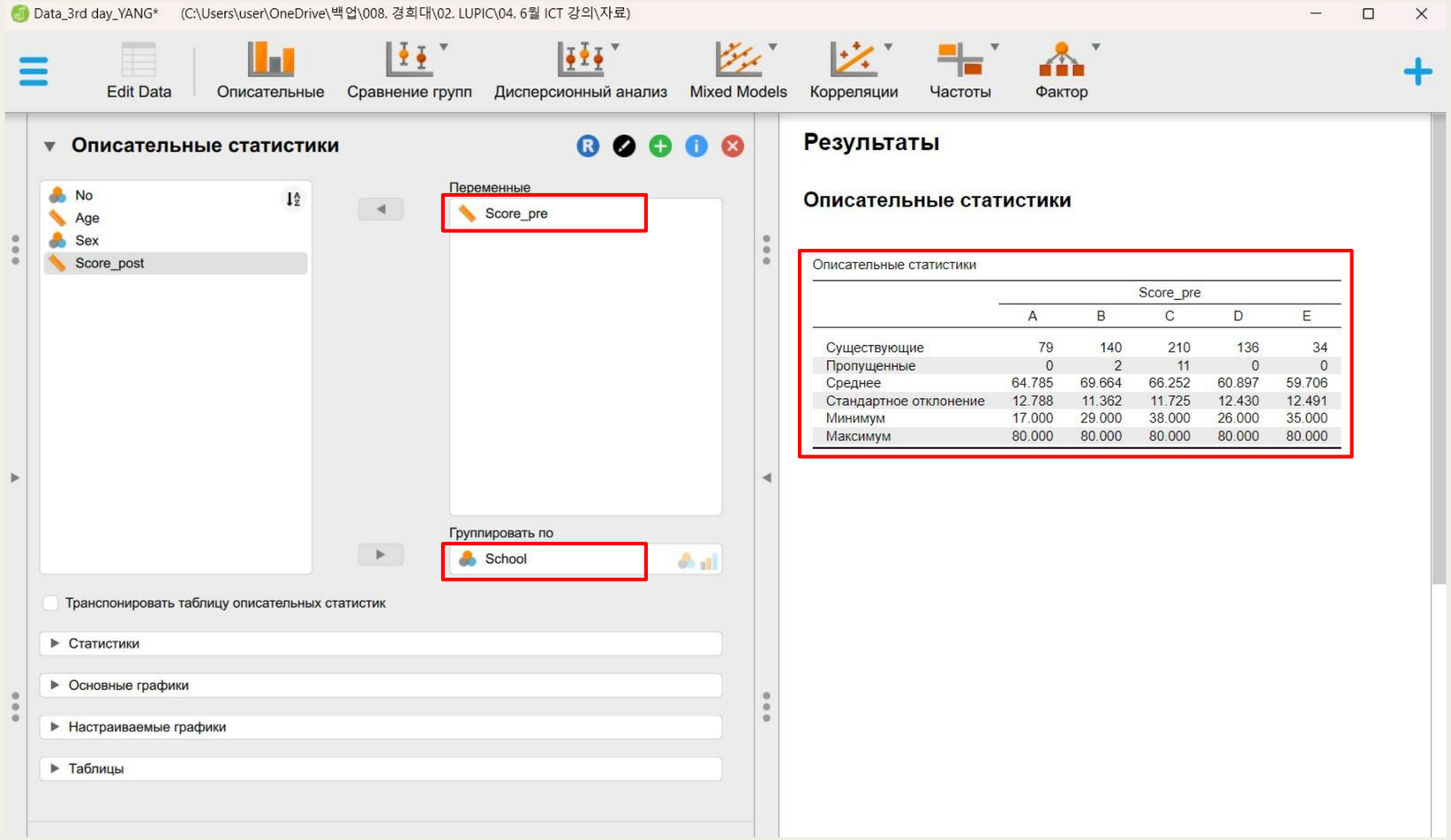

Описательные статистики

	Score_pre				
	A	B	C	D	E
Существующие	79	140	210	136	34
Пропущенные	0	2	11	0	0
Среднее	64.785	69.664	66.252	60.897	59.706
Стандартное отклонение	12.788	11.362	11.725	12.430	12.491
Минимум	17.000	29.000	38.000	26.000	35.000
Максимум	80.000	80.000	80.000	80.000	80.000

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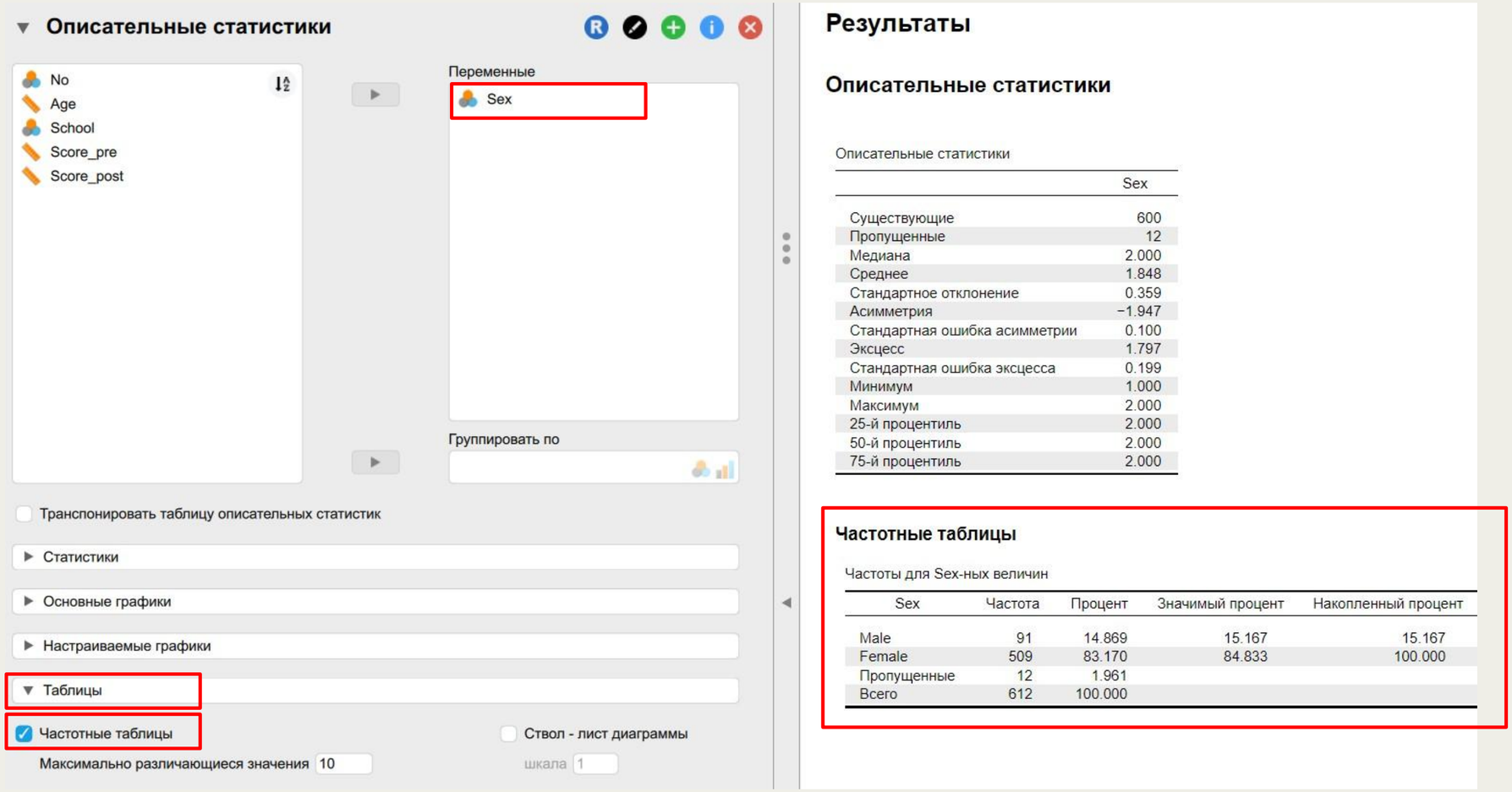
6. Descriptive



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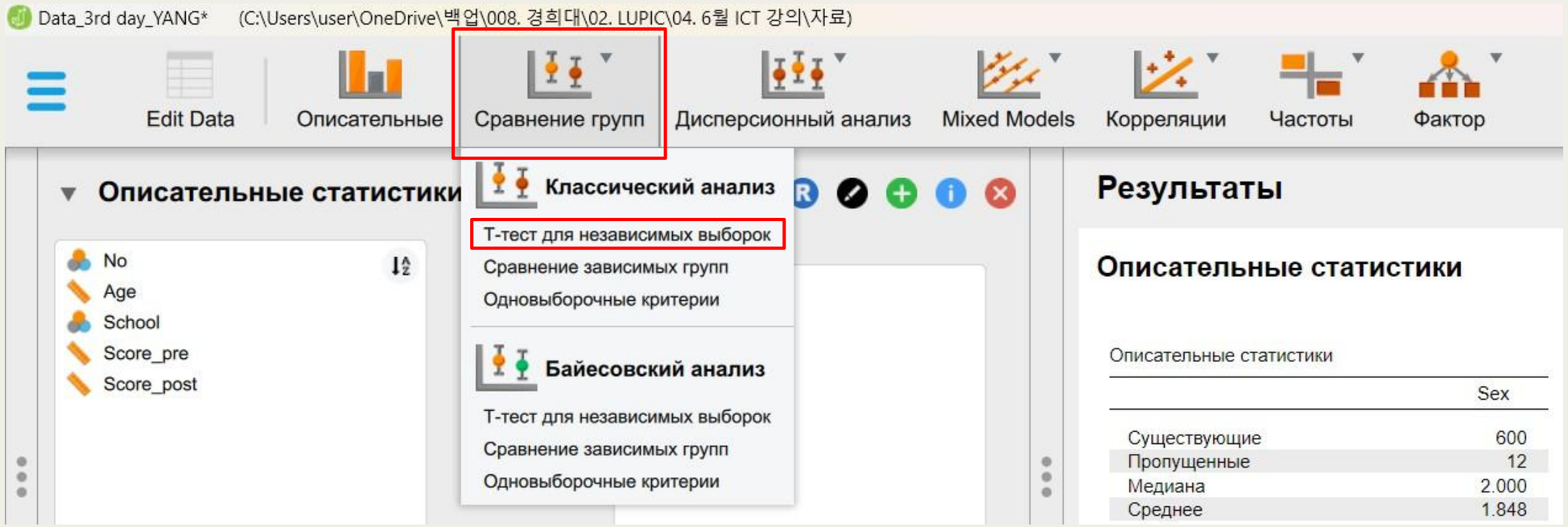
7. Two Group Comparison



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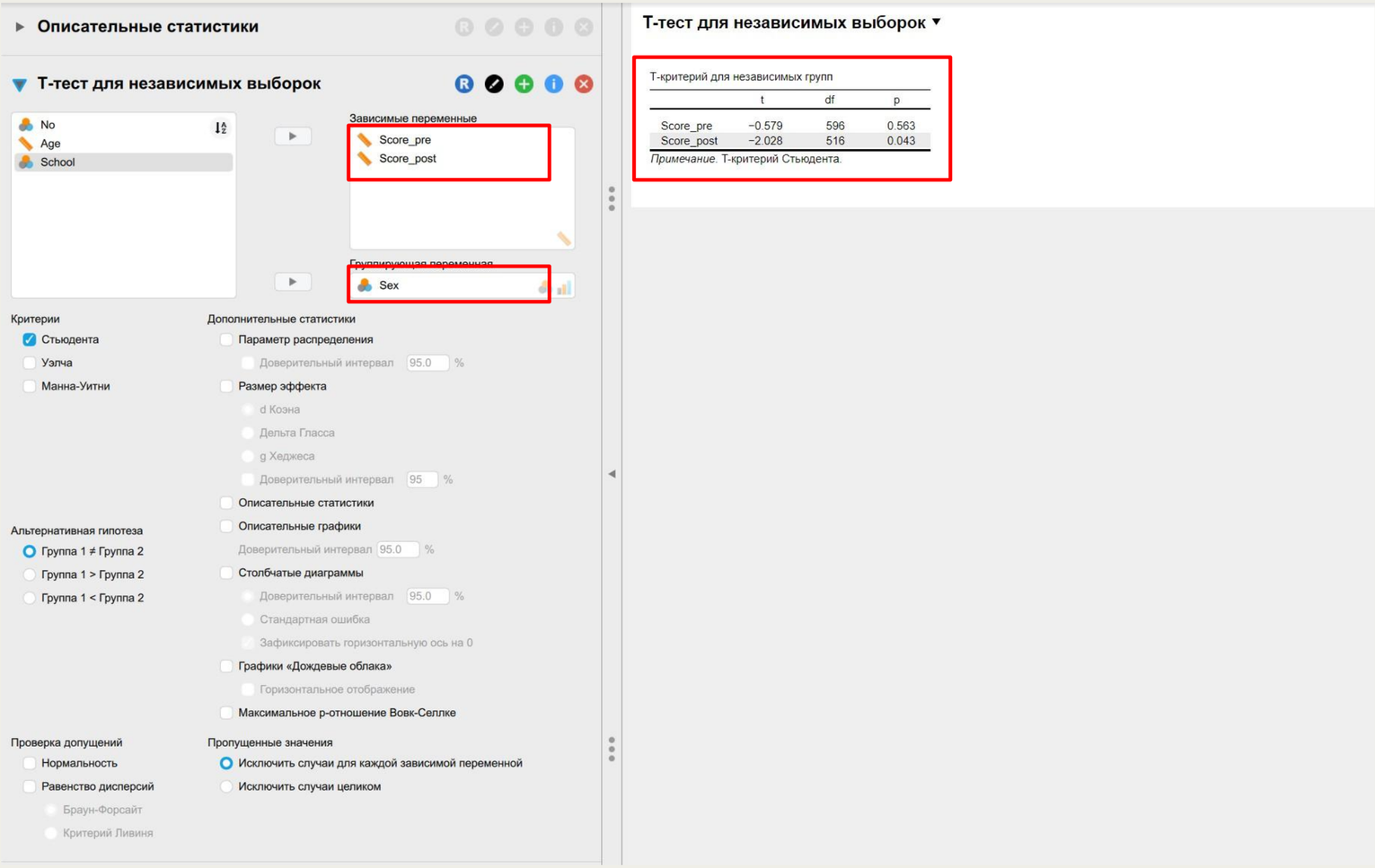
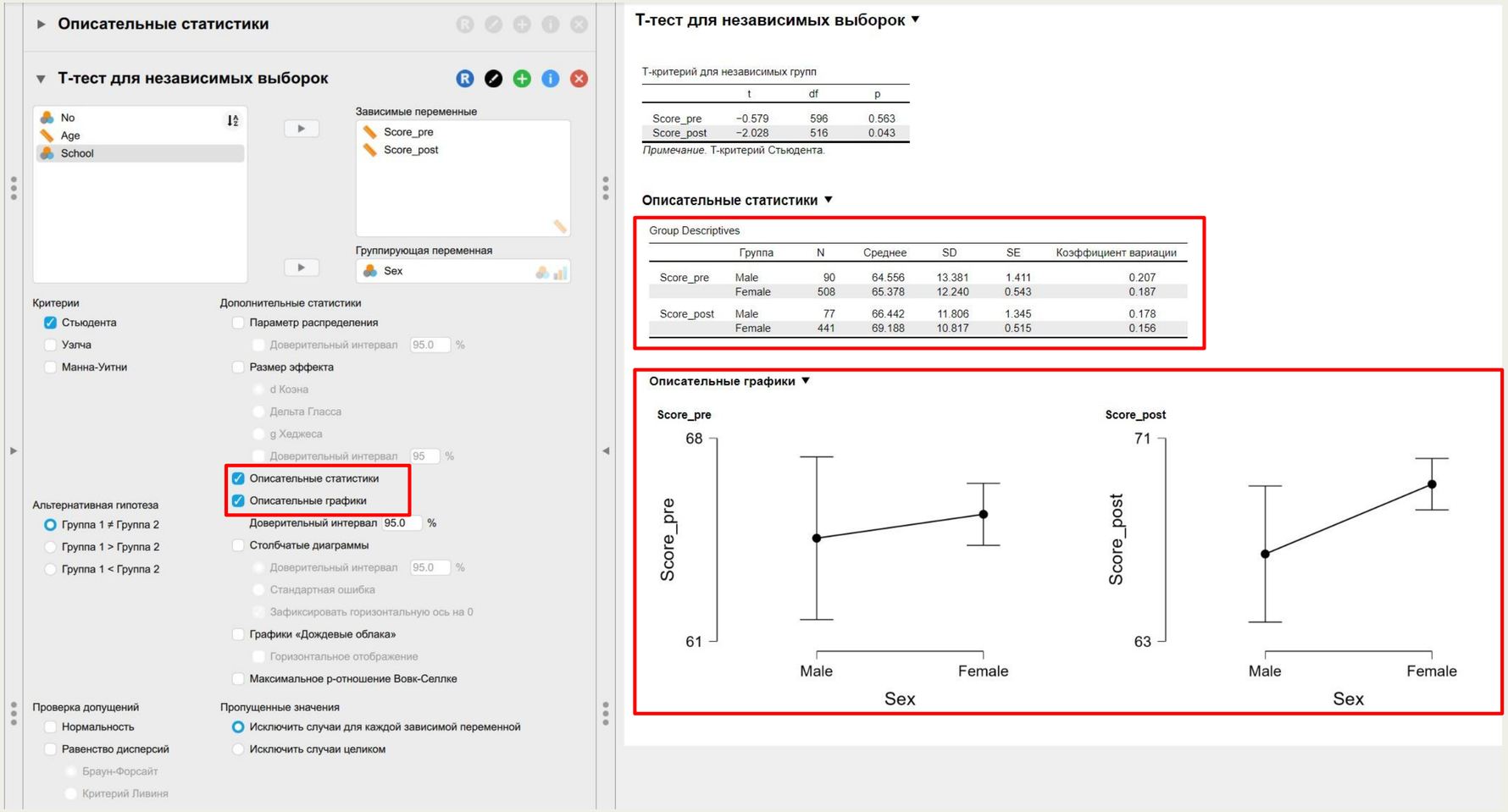


7. Two Group Comparison



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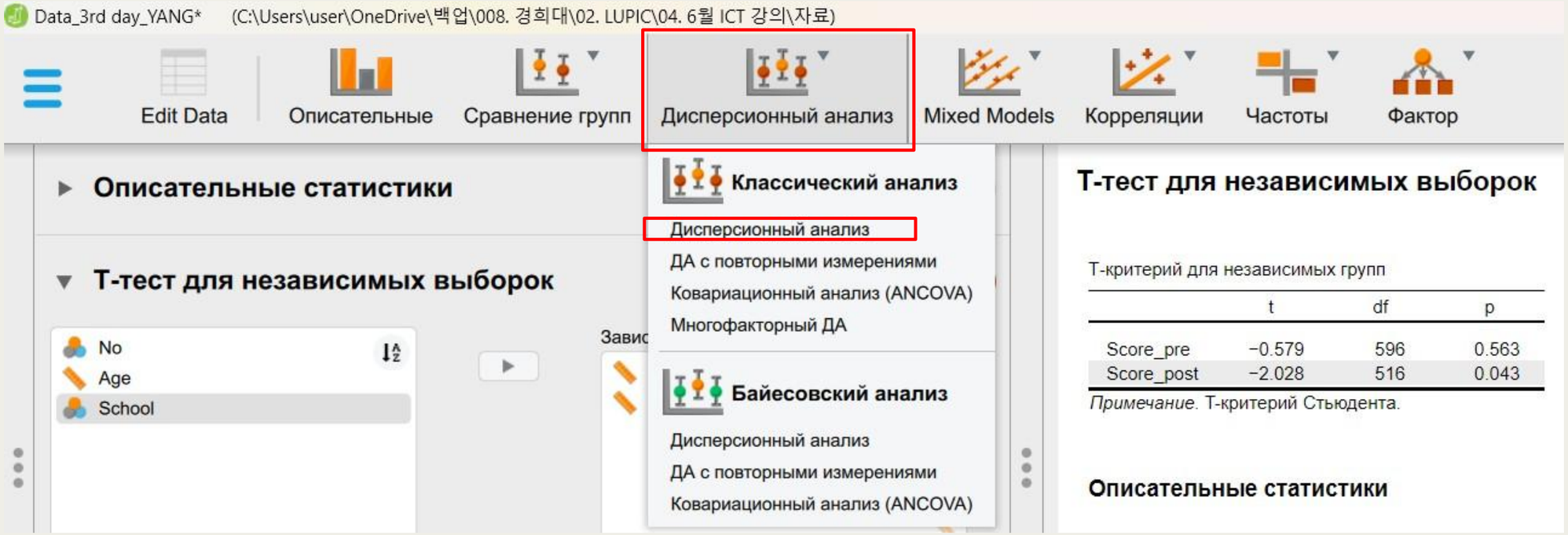
8. ≥ 3 Group Comparison



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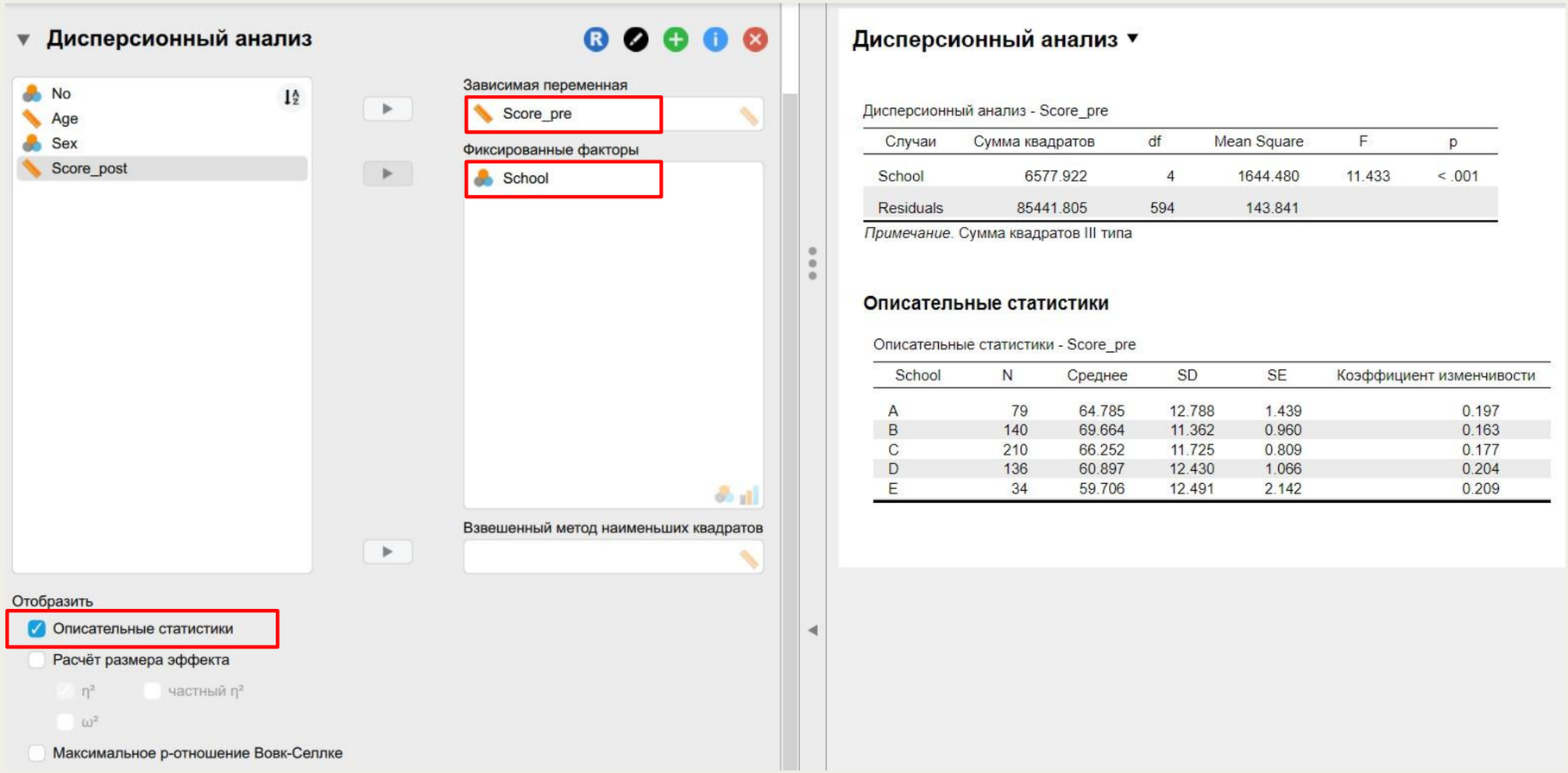
8. ≥ 3 Group Comparison



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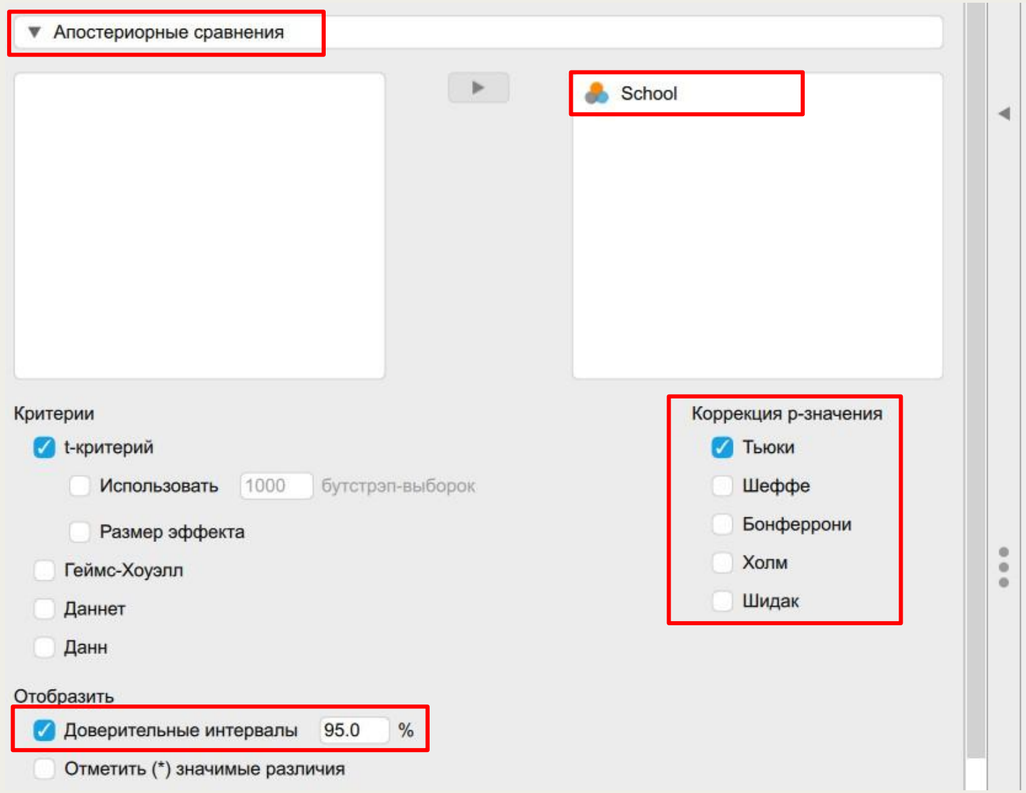
8. ≥ 3 Group Comparison



Апостериорные сравнения

Т-критерий

Апостериорные сравнения - School

			95% CI for Mean Difference				
		Разница средних	Нижняя граница	Верхняя граница	SE	t	p_{tukey}
A	B	−4.879	−9.497	−0.262	1.688	−2.891	0.032
	C	−1.468	−5.799	2.864	1.583	−0.927	0.886
	D	3.888	−0.754	8.530	1.697	2.292	0.149
	E	5.079	−1.652	11.810	2.460	2.065	0.237
B	C	3.412	−0.169	6.992	1.309	2.607	0.070
	D	8.767	4.816	12.718	1.444	6.072	< .001
	E	9.958	3.684	16.233	2.293	4.343	< .001
C	D	5.355	1.743	8.967	1.320	4.057	< .001
	E	6.546	0.480	12.613	2.217	2.953	0.027
D	E	1.191	−5.101	7.483	2.300	0.518	0.986

Примечание. P-value and confidence intervals adjusted for comparing a family of 5 estimates (confidence intervals corrected using the tukey method).

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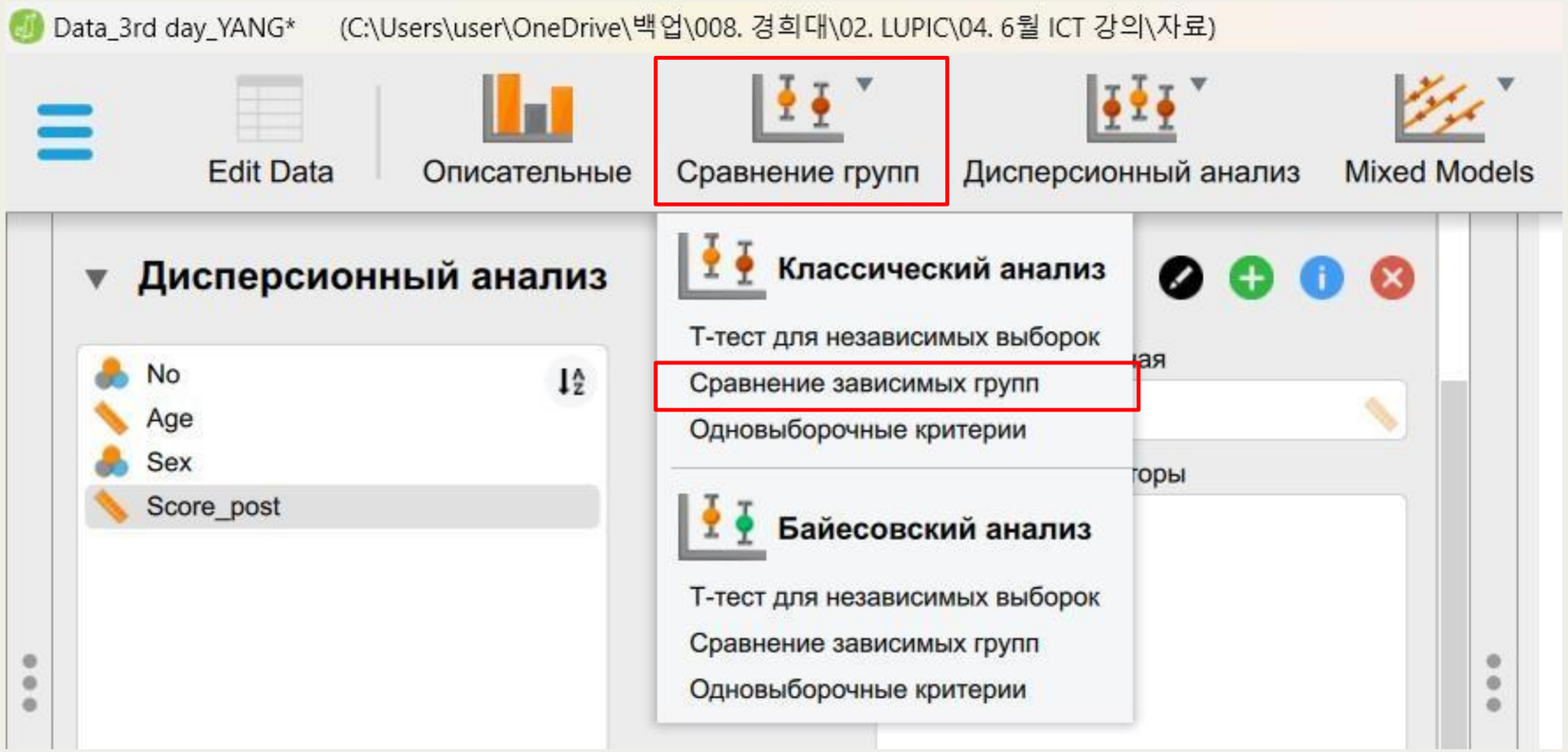
9. Paired Sample T-Test



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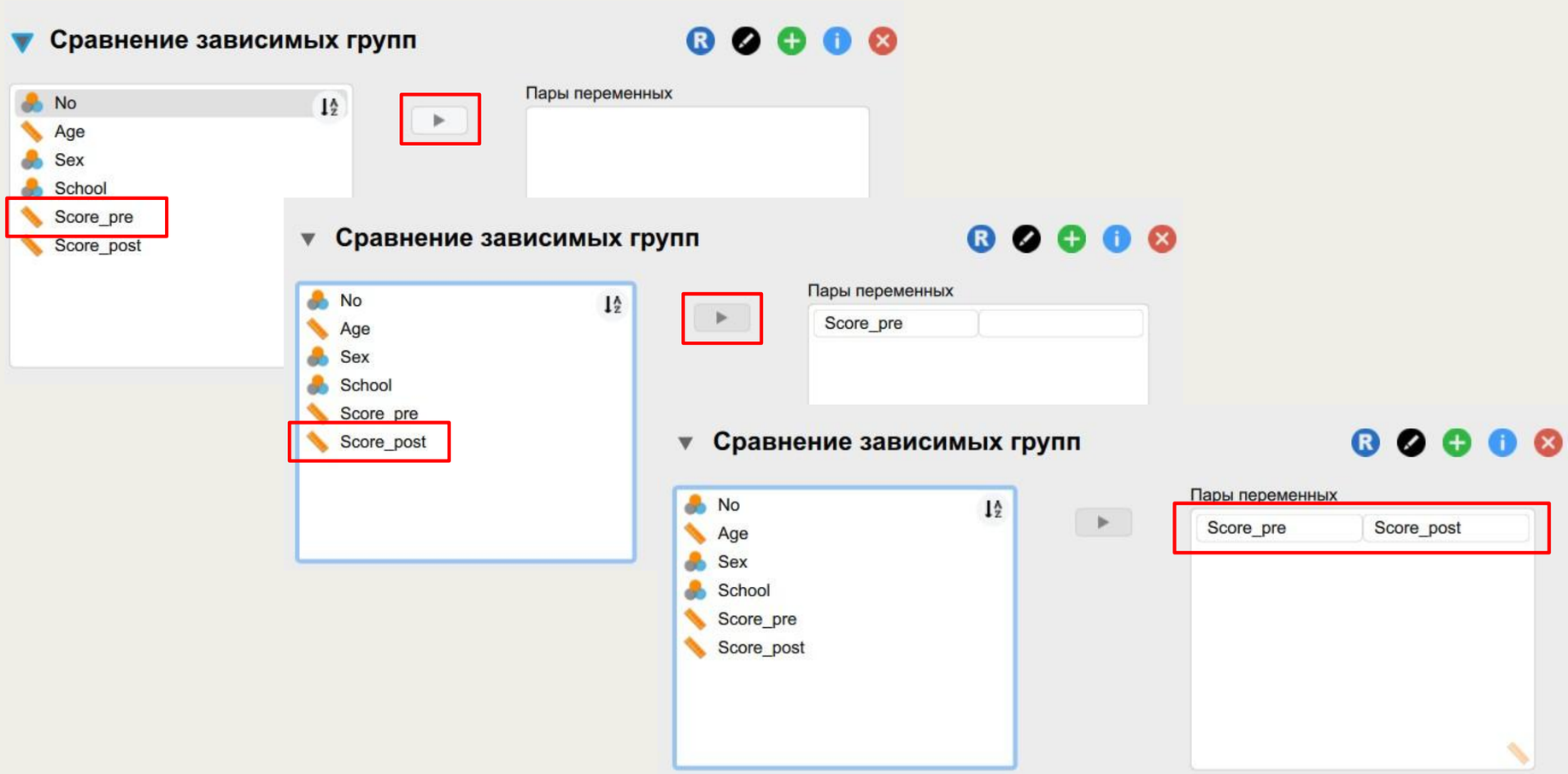
9. Paired Sample T-Test



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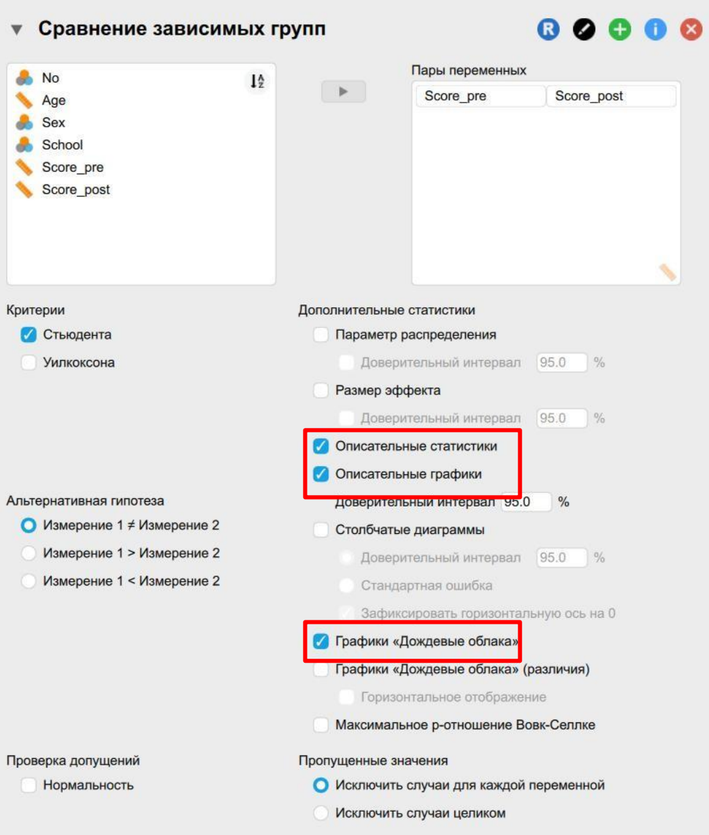
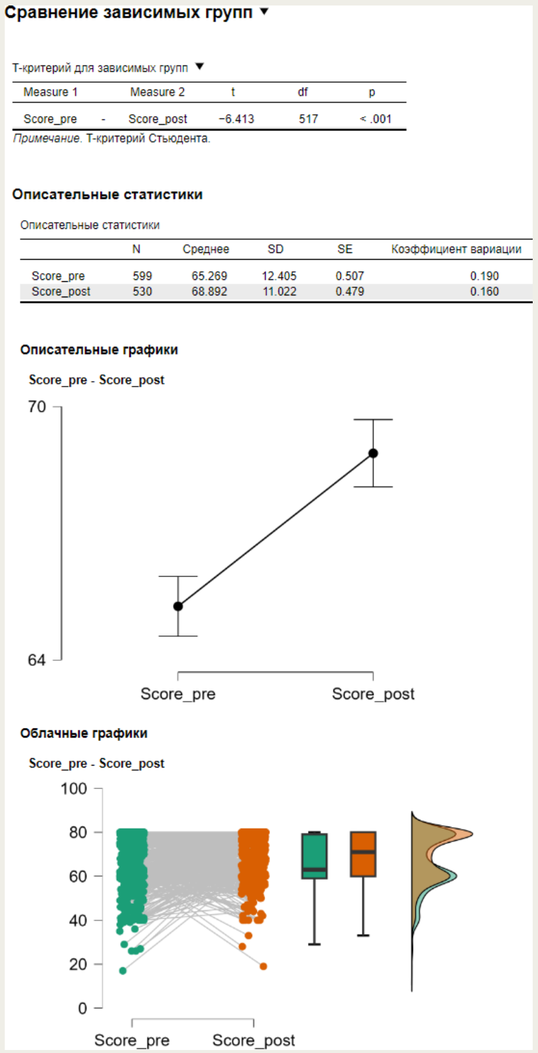
9. Paired Sample T-Test





Сравнение зависимых групп

Т-критерий для зависимых групп

Measure 1		Measure 2	t	df	p
Score_pre	-	Score_post	−6.413	517	< .001

Примечание. Т-критерий Стьюдента.

Описательные статистики

Описательные статистики

	N	Среднее	SD	SE	Коэффициент вариации
Score_pre	599	65.269	12.405	0.507	0.190
Score_post	530	68.892	11.022	0.479	0.160

Описательные графики

Облачные графики

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9. Paired Sample T-Test

성별 또는 학교별로 나눠서 보고 싶다면?

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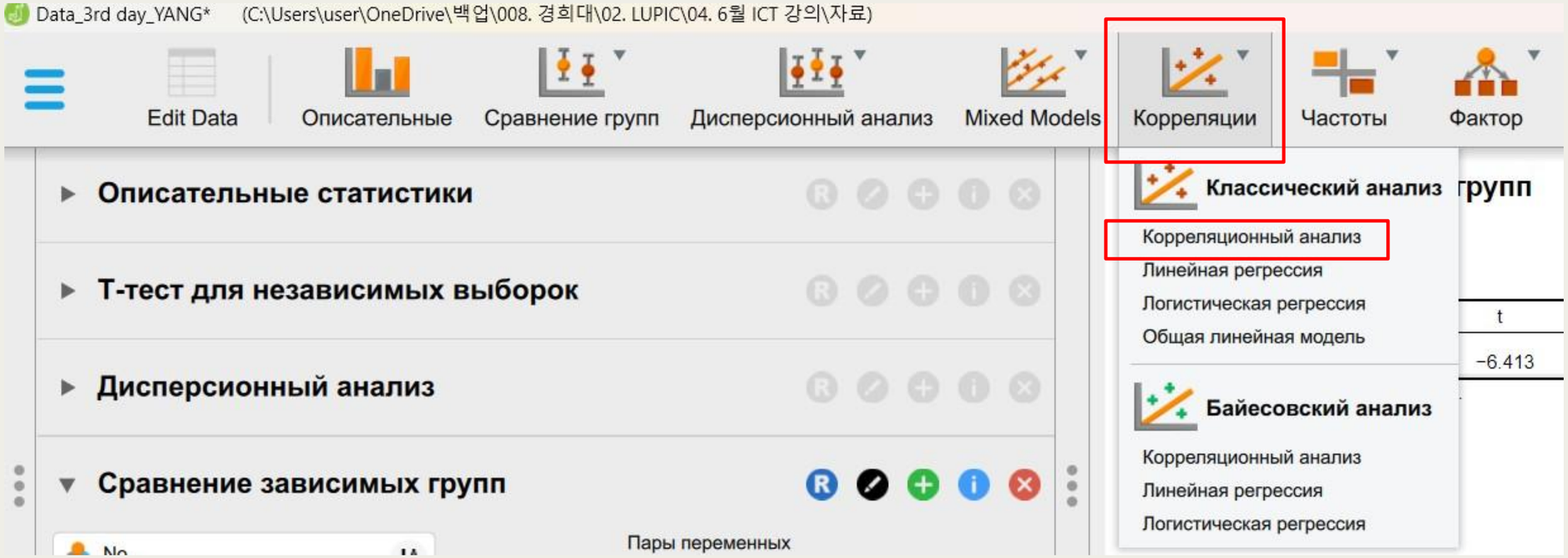
10. Correlation



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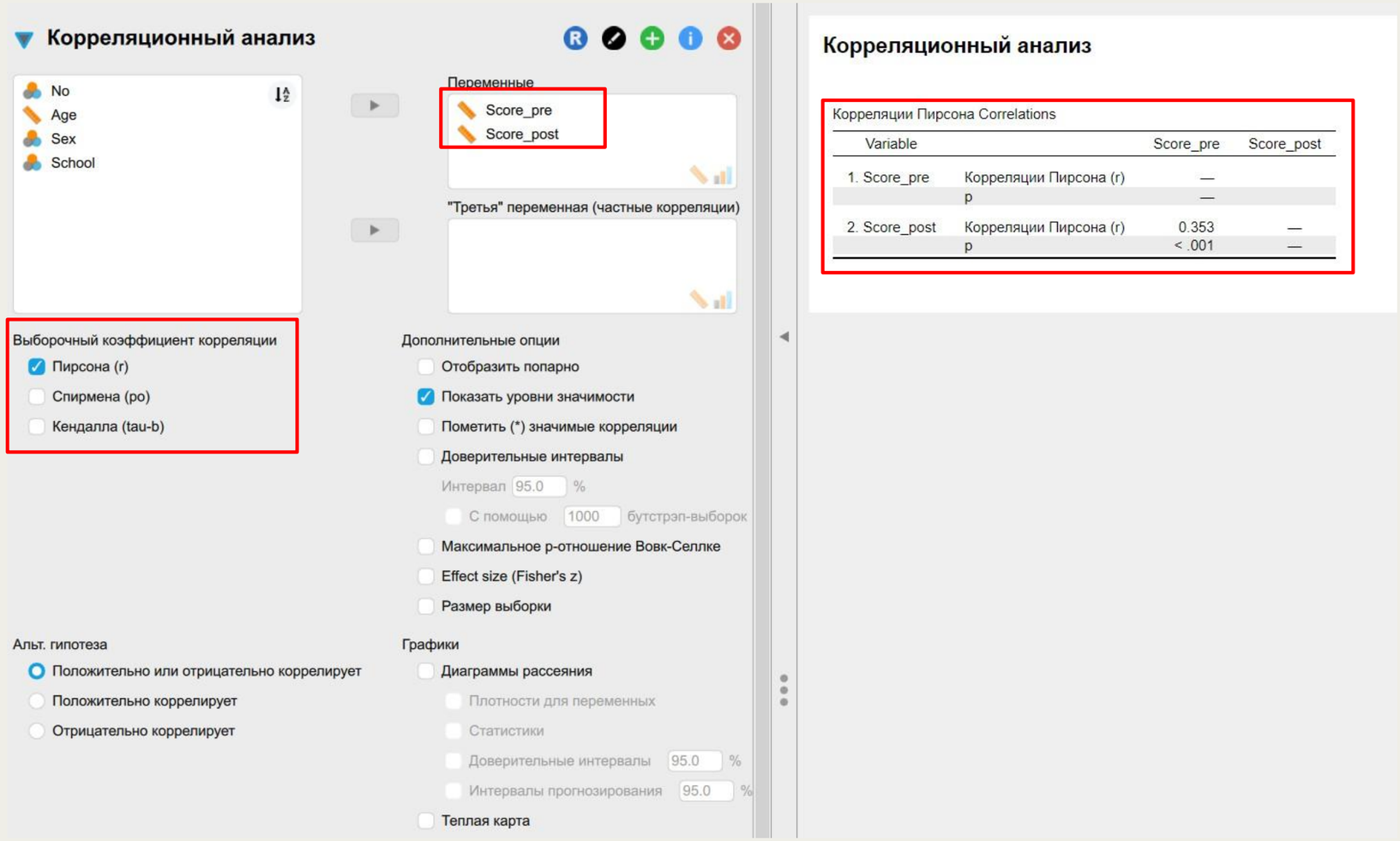
10. Correlation



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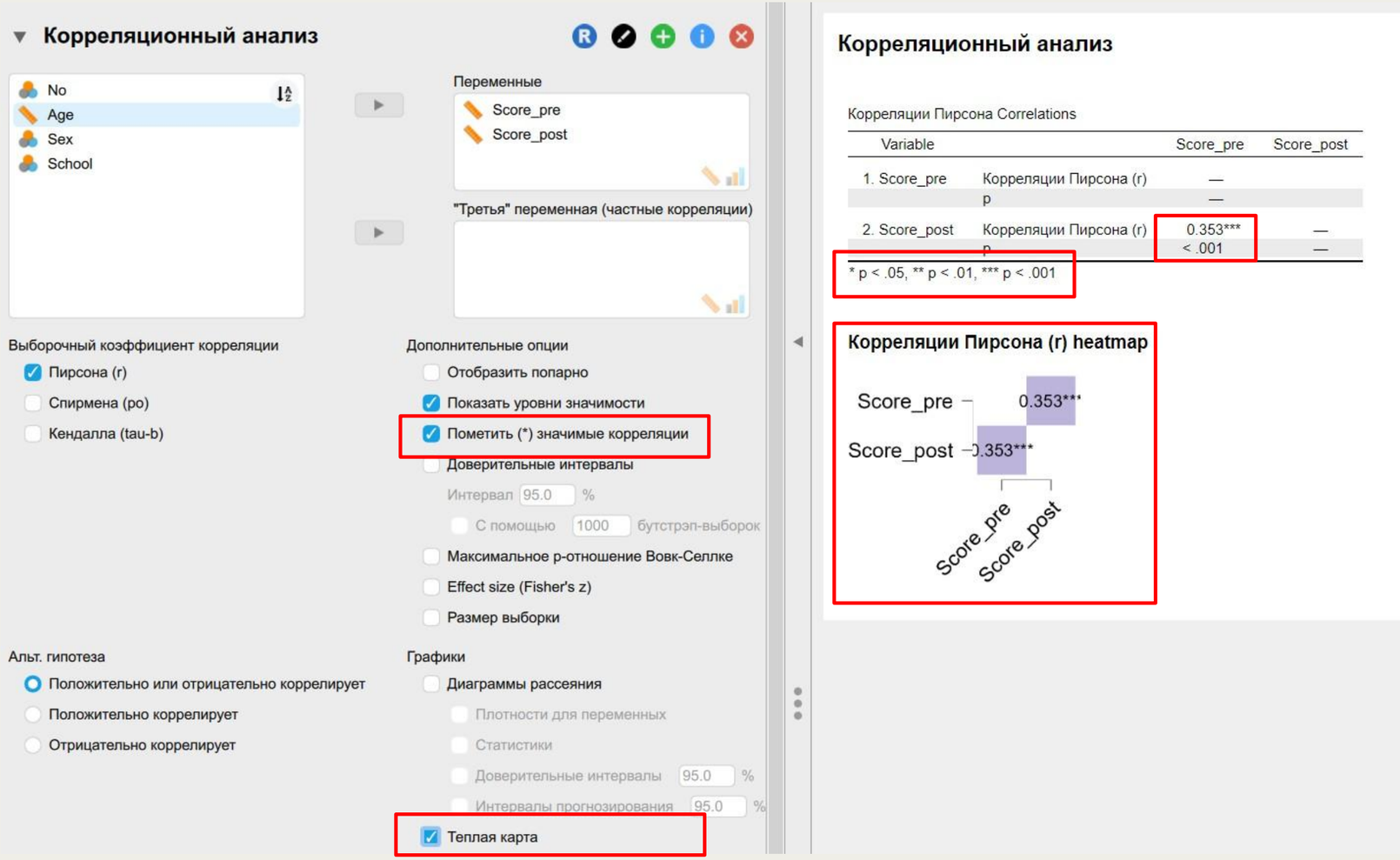
10. Correlation



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The End

3rd day (2024/06/26) Faculty Development Program

Seung Hyeon Yang
Research Professor, College of Nursing Science, Kyung Hee University, Republic of Korea

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